

Intrinsic-Dimension-Guided Bottleneck Selection for Variational Autoencoders with Active-Unit Diagnostics

Chika Obata ¹ , Mizuki Dai ¹ , and Kenya Jin'no ^{1,*} 

¹ Department of Intelligent Systems, Faculty of Information Engineering, Tokyo City University, 1-28-1 Tamazutsumi, Setagaya-ku, Tokyo 158-8557, Japan; g2332016@tcu.ac.jp (C.O.); g2591402@tcu.ac.jp (M.D.)

* Correspondence: kjinno@tcu.ac.jp; Tel.: +81-3-5707-0104 (K.J.)

Abstract

Intrinsic-dimension estimates and active-unit diagnostics address different aspects of variational-autoencoder bottleneck selection. We evaluate search policies on four image datasets, five training seeds and five regularisation weights, using a validation quality target and common recorded candidate costs. A start-width ablation holds the reliability checks, fallback and reference-bank charges fixed. The ID-derived and midpoint starts return identical widths in all 100 conditions; where ID is accepted, the former requires 8.84 rather than 5.84 candidates on average. Two errors of an ungated midpoint policy occur in conditions where the ID policy instead uses ascending fallback. Across 10,000 retrospective coefficient and threshold settings, 155 local outputs miss the smallest qualifying width. Additional three-seed experiments on each dataset evaluate ARM-based pruning and exact-Jacobian relevance readouts. Native reconstruction, dimension transfer and constraint satisfaction give distinct outcomes; dSprites remains problematic for the local ARM implementation. A separate one-seed MNIST experiment compares Monte Carlo ELBO capacity selection. Active-unit prediction does not establish additional benefit on its near-ceiling task. These results identify how search rules, implementation choices and evaluation definitions affect measured quality and cost, without establishing universal dimension-selection guarantees.

Keywords: variational autoencoder; bottleneck dimension; intrinsic dimension; active units; model selection; reconstruction quality; reproducibility

1. Introduction

The latent width of a variational autoencoder (VAE) [1] controls the capacity available to its representation. Choosing that width by validation over a candidate grid is straightforward, but requires training several models. Alternatives include intrinsic-dimension (ID) estimation [2,3], comparisons between sampled and mean latent representations [4], activity diagnostics [5,6], and pruning through reconstruction constraints or relevance priors [7,8]. These signals answer different questions: geometric dimension, coordinate use, reconstruction quality and network width need not coincide.

This study asks how the use of ID changes the decisions and training requests of a latent-width search, whether AU diagnostics add predictive value beyond other recorded quantities, and how these measurements depend on quality targets and readout settings. We compare policies on common candidate-model families and fixed splits. Search order, stopping rules, quality attainment and agreement with the smallest qualifying grid width are measured separately.

Received:

Revised:

Accepted:

Published:

Copyright: © 2026 by the authors.

Submitted to *Mach. Learn. Knowl. Extr.* for possible open access publication under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

The contribution is an empirical evaluation of ID-guided bottleneck selection and AU diagnostics under explicit quality and cost definitions. Four image datasets, five candidate seeds and five regularisation weights are covered. A start-width ablation holds reference acquisition, reliability checks and fallback fixed, isolating the effect of the ID-derived start from the effect of the search rule. A factorial analysis examines the ID coefficient and reliability thresholds.

Additional three-seed experiments on each dataset assess an ARM-based pruning implementation and exact-Jacobian relevance readouts. Native model quality, dimension transfer and training-constraint satisfaction are reported separately. Component checks improve the implementation audit but do not establish full reproduction of the original pruning methods. A separate MNIST experiment evaluates Monte Carlo ELBO capacity selection. Sections 3–5 define the protocols, Section 6 reports results, and Section 7 states the scope of inference.

2. Related Work

2.1. Dimension, Reconstruction and Latent Activity

PCA [9], Isomap [10] and locally linear embedding [11] construct low-dimensional representations. ID estimators instead quantify local dimension from sample geometry. We use TwoNN and an MLE cross-check; other approaches include DANCo, expected simplex skewness and geodesic entropic graphs [12–14]. Estimator limitations and dependence on the sampled distribution motivate cross-checks [15,16]. Graph-based methods are outside this experiment’s scope; we do not infer that they would resolve or reproduce the observed disagreements. Minimal-spanning-tree isolation forests also use graph geometry for anomaly scoring [17]; that objective is distinct from selecting a VAE width under a reconstruction target. Lower ID in learned representations has also been observed in deep networks [18,19].

FONDUE [4] searches for the capacity preceding a separation between ID estimates of sampled and posterior-mean representations. Its appendix gives FONDUE-VAR, which chooses a count from latent-variable types. AU and variable types share an interest in unused coordinates but are different statistics: AU thresholds the across-data variance of posterior means, whereas the type classifier uses moments of posterior variances [6]. Agreement of their final counts is not equivalence of their stability or information content.

GECO with L_0 regularisation optimises a gated VAE under a reconstruction constraint [7,20]. Its source uses ARM; the historical local adaptation used hard-concrete (HC) gates [21]. The additional ARM implementation restores this gradient component but retains differences in initialisation, constraint scheduling and KL scaling (Section 5.5). Our validation anchor supplies an external quality yardstick.

ARD-VAE uses a hierarchical relevance prior [8]. The source tunes a regularisation weight, uses a Gaussian approximation for its closed-form KL, and selects axes explaining 99% of a Jacobian-weighted variance score. The Gaussian approximation is thus shared with the source. Historical finite-step scores and their normalisation audit appear in Appendix C.2; the new validation uses automatic-differentiation Jacobians. We distinguish the source’s zero-centre variance update from an additional sample-centred variance-update variant, and retain local training and sampling settings.

2.2. Relation to the Conference Paper and Earlier Manuscript

Our NOLTA2025 contribution [22] concerned the relation between input ID and latent-space capacity. The available conference manuscript describes class-specific MNIST experiments with four fully connected and two convolutional architectures and four seeds. CNNs and multiple seeds are consequently not new features of this revision. The conference

manuscript available for comparison has a working title different from the published title; the reference uses the title and pages listed by the authors' institutional publication record. Table 1 distinguishes this earlier scope from the present comparison. MAKE47 denotes the earlier journal submission, not a separate publication.

Table 1. Scope of the conference study, earlier journal submission and present revision.

Item	NOLTA2025	MAKE47	Present revision
Question	Input and latent ID versus capacity	ID-guided choice with AU verification	Selection quality, cost and diagnostic limitations
Models	Four FC and two CNN structures; MNIST classes; four seeds	FC/CNN and synthetic/image extensions	Fixed candidate families on four image datasets; five candidate seeds
Comparators	Architecture and capacity comparisons	Capacity sweeps and diagnostic checks	FONDUE variants, grid criteria and search policies; separate pruning validation
Readouts	ID estimates	AU-based operational checks	Same-checkpoint type/AU trajectories and paired predictive analysis
Evidence	Conference settings	Earlier loss convention	Corrected-loss records; explicit targets, final test metrics and accounting limits

3. Evaluation Framework

3.1. Models and Quality Target

We distinguish the *ID-reference AE*, which supplies a representation for estimation, the *quality anchor*, an ordinary VAE at the largest candidate width, and the *selected model*, the ordinary VAE evaluated after dimension selection. Write $\mathcal{M}$ for the candidate grid, s for a training seed, and $D_s(m, \beta)$ for validation per-pixel MSE of the deterministic reconstruction $g(\mu(x))$. The implemented target is

$$m_a = \max \mathcal{M}, \quad T_{D,s}(\beta) = D_s(m_a, \beta) + \max\{0.10D_s(m_a, \beta), 0.001 \text{Var}(X_{\text{val}})\}. \quad (1)$$

The variance is computed over all validation-array entries. The relative term dominates in every recorded image condition, giving $T_{D,s} = 1.10D_s(m_a, \beta)$. The configuration originally also specified a three-seed-SD term, but the executed target function was supplied one anchor at a time and omitted that term. We report the implemented rule; we do not retrospectively describe it as a pooled-seed target. Appendix A lists every dataset- β anchor and target.

For a fixed seed and β , the target is held constant during search. It is recomputed from a different anchor when β changes. A β -sweep therefore changes both the learned representation and the quality yardstick. It is not a fixed-quality intervention. A method that returns a valid width need not meet this external target; quality attainment is evaluated separately for every final model.

3.2. Canonical Replay Prices and Historical Ledgers

Replay uses one immutable candidate snapshot. A logical key contains dataset, width, training seed, requested epochs, early-stopping option and β ; architecture, likelihood and split are fixed by the dataset configuration. Each unique requested key, including the quality anchor, is charged its duration in this snapshot once. Metrics and prices are

therefore identical whenever replay policies request the same key. These are paired uses of stored records, not independent retraining.

Earlier method ledgers sometimes attach different durations to the same logical key. Even when validation and test summaries agree exactly, missing historical weights and execution identifiers prevent proving checkpoint identity. We therefore do not mix those prices in the new policy comparison. Appendix B.1 preserves the historical account and resolves every anchor/final role to its key, source and both available durations in a supplementary CSV. The quality anchor and its price in each replay come from the same record.

Let C_{cand} sum the canonical prices of requested candidates and C_{ref} be the nine-AE bank's recorded training cost. Cold use charges $C_{\text{cand}} + C_{\text{ref}}$ for an ID policy; warm use charges C_{cand} when that bank already exists. A 25-task batch (five seeds by five weights for one dataset) allocates $C_{\text{ref}}/25$ per task. This does not turn shared-bank outcomes into independent ID replications. All accounts exclude missing ID-computation and some post-training evaluation times. They estimate attributed training costs, not new end-to-end wall-clock measurements. Historical probe and FONDUE calibration costs are reported separately from the replay policies.

3.3. Outputs and Termination

Each trial records a raw returned dimension, the width of the evaluated model, the ID reliability decision where applicable, fallback use, the logged termination state, the budget flag, quality attainment, and whether the final width is smaller than the anchor. These events can overlap. In particular, fallback can produce a compact model that meets the target. The label SUCCESS in an older log indicates ordinary termination; it is not itself evidence of target attainment.

A budget-exceeded trial contributes its consumed time and, if it has a returned model, that model's dimension and metrics. A raw dimension below one is reported as degenerate; its time and failure count remain, while MSE is missing because no final model exists. The original implementation checks budgets after completed calls and can still obtain a final model. The reported limits are therefore soft accounting limits, not hard interruptions. Outcome tables preserve these distinctions.

3.4. Reference Bank and Reliability Checks

The reference bank uses three widths and three fixed reference seeds, shared across the five candidate seeds. Let $\bar{d}_r$ be the mean TwoNN estimate at reference width r , $\hat{d} = \text{mean}_r \bar{d}_r$, and $\bar{d}_{\text{MLE}}$ the mean MLE estimate over that same bank. The checks are

$$R = \frac{\max_r \bar{d}_r - \min_r \bar{d}_r}{\text{median}_r \bar{d}_r} \leq 0.15, \quad A = \frac{|\hat{d} - \bar{d}_{\text{MLE}}|}{\hat{d}} \leq 0.25. \quad (2)$$

The values $c = 2$, $R_{\text{max}} = 0.15$ and $A_{\text{max}} = 0.25$ were internal heuristics recorded in the 12–13 September specification. The factor two was geometrically motivated but not derived for this statistical task; neither reliability tolerance was calibrated to a false-rejection rate. Their numerical roles are examined retrospectively rather than justified by the quality-margin experiment.

3.5. Retrospective Search Policies

We define the following replay on 19 September, after the earlier results and review. A full-grid reference returns $m^* = \min\{m \in \mathcal{M} : D_s(m, \beta) \leq T_{D,s}\}$. Ascending search obtains the anchor and visits increasing widths until the first pass; it identifies m^* without a monotonic-error assumption. The historical ID procedure exhausts a lower window

when ID is accepted and uses minimum-MSE coarse search otherwise. Its full executed pseudocode is retained in Appendix C.1.

The revised *ID-local* policy uses ID to choose a starting width rather than an exhaustive window. If the start passes, it walks downwards to the first failure; otherwise it walks upwards to the first pass. This can omit many candidates, but can miss a smaller qualifying width if the pass/fail sequence is not monotone. Rejected ID banks now invoke ascending- Q fallback, not the historical coarse-MSE fallback. This fallback change is part of the retrospective policy.

Algorithm 1 (ID-local replay and start-width controls).

Input: grid, seed, weight, validation records and canonical prices, reference bank, c , $R_{\max}$, $A_{\max}$; no test metrics.

1. Request the largest width once; compute its target from Equation 1. A pass means validation MSE at most that target.
2. For ID-local and ID-gated midpoint, charge the reference bank and compute Equation 2. If either threshold fails, request ascending widths and return the first pass. Bank-free midpoint skips this step.
3. For accepted ID-local, start at the smallest grid width at least $c\hat{d}$, clipped to the largest width. For either midpoint policy, start at zero-based index $\lfloor (|\mathcal{M}| - 1)/2 \rfloor$.
4. If the start fails, request larger widths in order and return the first pass. The anchor guarantees a returned qualifying width.
5. If the start passes, request adjacent smaller widths in descending order until one fails, or the smallest width is reached; return the last pass.
6. For the optional certified ID variant, test previously unqueried widths below the local answer in ascending order, stopping at the first pass. This certifies the full-grid minimum without monotonicity.
7. Charge each distinct candidate once. Return width, requested keys, count, reference/candidate costs and the soft-budget flag. Attach test metrics only after the selection is fixed.

Full-grid- Q , ascending- Q , the legacy path, ID-local, bank-free midpoint-local and certified ID-local are replayed on all 100 dataset-weight-seed conditions. The additional ID-gated midpoint control adds 100 replays, for seven policies and 700 rows in total. Paths are completed within the stored grid (at most 12 distinct candidates); the original time limit is recorded as a flag rather than interrupting this counterfactual evaluation. Legacy rows retain their executed path and output, including original fallback. We report quality, the width gap $m - m^*$, exact agreement, candidate requests and canonical cost. The bank-free midpoint also removes the reliability gate and its fallback, so its comparison with ID-local does not isolate the starting width. ID-gated midpoint holds these components and their charges fixed and changes only the accepted starting width. Midpoint is a fixed control, not a tuned or universally optimal start.

The coefficient set is $\{1, 1.5, 2, 3, 4\}$, inherited from the earlier configuration. The retrospective reliability grids are $R_{\max} \in \{0.10, 0.15, 0.25, 0.50\}$ and $A_{\max} \in \{0.15, 0.25, 0.40, 0.85, 2.00\}$. They include stricter settings and settings on either side of the observed dataset decisions. The most permissive values are stress tests, not recommended defaults. All 10,000 combinations of these parameters with the 100 recorded conditions are retained; no best-performing threshold replaces the original default.

4. Theoretical Background

For an ideal smooth d -manifold, Whitney's embedding theorem gives an embedding into $\mathbb{R}^{2d}$ [23]. Exact continuous reconstruction through an injective encoder entails an embedding, but neither a noisy finite-sample ID estimate nor an approximate neural reconstruction satisfies these assumptions automatically. The multiplier two is therefore

a heuristic under evaluation, not a quality guarantee or a lower bound inferred from measured ID.

In linear Gaussian latent-variable models, informative directions depend on the covariance spectrum and noise scale; analyses of linear VAEs and posterior collapse make this relation explicit [9,24]. VAE alignment with principal directions has also been studied [25]. These results motivate measuring how regularisation affects coordinate use. They do not equate AU with intrinsic or minimum embedding dimension in a nonlinear VAE. AU also depends on the latent coordinates and the threshold used to count them.

5. Experimental Setup

5.1. Data, Architecture and Optimisation

Table 2 gives the actual image-data settings. MNIST [26], Fashion-MNIST [27] and CIFAR-10 [28] use random subsets of the official training and test pools. A permutation with seed 20260912 selects 20,000 training-pool examples; another with seed 20260913 divides them 90:10. The 3000 test examples use seed 20260914. For dSprites [29], which has no official train/test split, the first 20,000 and next 3000 entries of one permutation of 737,280 images are disjoint pools; the 20,000 are divided 90:10 and each index list is sorted before loading. Images are scaled to $[0, 1]$, with no augmentation. Training seeds are 42, 123, 777, 2024, 31337; the split does not vary with these seeds.

Table 2. Image-data settings used for candidate VAEs. All datasets use 18,000 training, 2000 validation and 3000 test examples. G: Gaussian; B: Bernoulli. Budgets are recorded-time limits with a common limit of 12 candidate-training calls; ID AEs are additional.

Dataset	Input	Likelihood	Candidate widths	Budget (s)
MNIST	$1 \times 28 \times 28$	G	2, 4, 6, 8, 10, 12, 16, 20, 24, 32, 48, 64	1470
Fashion-MNIST	$1 \times 28 \times 28$	G	Same as MNIST	1470
dSprites	$1 \times 64 \times 64$	B	Same as MNIST	5163
CIFAR-10	$3 \times 32 \times 32$	G	8, 16, 24, 32, 48, 64, 96, 128, 192, 256	2924

For MNIST and Fashion-MNIST, the encoder has dense widths 512 and 256, ReLU activations, and separate linear mean/log-variance heads; the decoder has widths 256 and 512 with a linear output. The CIFAR-10 encoder has channels 32, 64 and 128; dSprites adds 256. Convolutions use kernel 4, stride 2, padding 1 and ReLU, leaving 4×4 feature maps, followed by linear posterior heads. A linear/ReLU projection and reversed transposed convolutions form the decoder, ending in a sigmoid. The ID-reference AE flattens the input and uses dense widths $512-256-r$ and the reverse decoder, with a linear output; it is separate from the candidate CNNs.

Ordinary candidates use Adam, learning rate 10^{-3} , default moments $(0.9, 0.999)$, $\epsilon = 10^{-8}$, no weight decay, batch size 128 and up to 300 epochs. The last batch has 80 examples (141 updates per epoch). ID AEs use the same optimiser and batch size for 100 epochs. Early stopping uses patience 30. A recorded plateau flag requires $(\max h - \min h) / |\text{mean } h| < 0.01$ over the last 50 validation-monitor values. It is not an additional stopping rule. Representative trajectory runs always complete 300 epochs.

The candidate training loss is

$$\mathcal{L}_\beta = \frac{1}{B} \sum_{i=1}^B \left[\ell(x_i, g(z_i)) + \beta \sum_{j=1}^m \frac{\mu_{ij}^2 + \exp(v_{ij}) - 1 - v_{ij}}{2} \right], \quad z_i \sim q_\phi(z | x_i), \quad (3)$$

where v is log variance. Gaussian reconstruction uses a sum of squared errors, equivalent to fixed observation variance $1/2$ up to an additive constant; dSprites uses summed binary cross-entropy. The primary weight is one; the ladder is $\{0.25, 0.5, 1, 2, 4\}$, constant during

each run. No cyclical annealing schedule [30] is used. Equal numerical β across likelihoods does not imply equal regularisation strength.

The stored validation monitor named “ELBO” is $h = -[\ell(x, g(\mu(x))) + \text{KL}]$, averaged over examples. It uses posterior-mean reconstruction, not an expectation over posterior samples, and is therefore an *ELBO proxy*, not an unbiased ELBO estimate or a certified bound. It also uses unit KL weight for checkpoint selection even when training uses another β . We retain these checkpoints and label the monitor accurately. No test metric enters the selection code. However, the candidate cache evaluates test metrics for every trained candidate, before method selection; the claim that only one final test evaluation was executed would be incorrect. Tables report test metrics only for the validation-selected models. This is a code-level separation of selection and evaluation, not a prospectively sealed test workflow.

5.2. Methods and Readouts

Grid search selects by validation MSE, the ELBO proxy, or linear-probe validation accuracy. Probes use posterior means, multinomial logistic regression, up to 2000 iterations, and $C \in \{0.01, 0.1, 1, 10\}$ chosen by validation. Labels are digit/apparel/ CIFAR classes and the three dSprites shapes. Both capacity and probe regularisation use validation labels in the probe selector. No centring, scaling or whitening is applied to the probe latents; logistic regression uses its default intercept. Coarse search uses Algorithm A1’s subset; the fixed baseline uses width 32.

FONDUE is a local PyTorch implementation using TwoNN and a fixed ID-gap threshold $0.2d_0$. It searches integer widths, which can lie outside the ordinary candidate grid. Its selected width is trained exactly. At $\beta = 1$, short training uses one epoch for MNIST and dSprites and two for Fashion-MNIST and CIFAR-10. dSprites follows the source’s epoch choice; elsewhere a seed-42 calibration increases epochs until two consecutive predictions agree, stopping at five. Equality is first tested after runs at one and two epochs; if predictions at $e - 1$ and e agree, the returned epoch count is $e - 1$, and both calibration costs remain charged. Other β -specific choices are in Appendix C. FONDUE+Q maps the returned width upwards onto the grid, obtains the anchor, and extends upwards until quality is met or its budget flag is set. This adds an external reconstruction rule to FONDUE.

FONDUE-VAR starts at round($2d_0$) and doubles width until the declared type condition is met. With mixed variables kept it returns active plus mixed once a passive variable appears; otherwise it returns the active count once a mixed or passive variable appears. We implement the description-based classifier: posterior variance is exponentiated *once* from log variance. A coordinate is passive if its across-data variance is below 0.1 and its mean is in $[0.9, 1.1]$, active if that variance is below 0.1 and its mean is at most 0.1, and mixed otherwise. The audited public-code version is identified in Appendix C.2; this is not claimed to reproduce every path of that code or its published experimental settings.

AU is independently defined by

$$\text{AU}_\delta = \sum_{j=1}^m \mathbf{1}\{\text{Var}_{x \in X_{\text{est}}} \mu_j(x) > \delta\}, \quad \delta \in \{10^{-3}, 10^{-2}, 10^{-1}\}, \quad (4)$$

with primary $\delta = 10^{-2}$. The across-example variances for AU, variable types and the target floor use denominator n (population variance); reported between-seed SDs use denominator $n - 1$. ID and activity readouts use the same 10,000 training examples sampled without replacement using seed 20260915. The MLE cross-check uses $k = 10$ neighbours. AU is diagnostic, not an extra selector in the method comparison.

The historical GECO and ARD adaptations start at the largest grid width and run 300 epochs. Their raw coordinate counts are mapped to the nearest candidate width, with

ties going to the smaller width, for the final ordinary VAE. Appendix C.2 specifies the local objectives, updates and readout differences. The evaluated MSE thus belongs to the transferred ordinary VAE, not to the original model.

5.3. Provenance and Scope of Prior Specification

The saved image-experiment provenance is Python 3.12.3, PyTorch 2.7.1+cu118, CUDA 11.8, cuDNN 90100 (9.1.0), NVIDIA driver 570.211.01, and an NVIDIA GeForce RTX 3070, on Linux 6.8.0-124-generic, x86-64, glibc 2.39. GPU provenance is recorded for these experiments; CPU numerical and probe operations accompany GPU training. The earlier MAKE47 manuscript did state PyTorch 2.0, RTX 3070 and a CPU fallback; it did not identify the complete CUDA/cuDNN/driver environment.

The configuration file was created on 12 September 2026, with subsequent changes recorded on 12–13 September. These are internal specification records, not a public preregistration. Calibration informed some changes, including the dSprites likelihood. The pruning additions and full ladder were run during 14–17 September. The paired random-feature comparison and first accounting audit were retrospective revisions dated 18 September. The search replay, factorial threshold analysis, normalized ARD readout and bounded model validation were added on 19 September. MNIST and other previously examined datasets are not described as wholly unseen scientific conditions. “Evaluation condition” in the predictive analysis means withheld from fitting that predictor.

The logged code token 8c2c6c28224b3450 hashes only the common module and configuration, not the entire experimental implementation; the earlier trajectory records use 99cb5eb7404ff6e7. A new SHA-256 file manifest identifies the inspected snapshot. Appendix G maps numerical results to their source files and describes remaining provenance limitations.

5.4. Bounded New Model Validation

The archived MAKE49 models are not available as compatible checkpoints. To evaluate an actual ELBO, we train a separate MNIST grid at $\beta = 1$, seed 42, on the same 18,000/2000/3000 split and the same architecture, optimizer, 300-epoch limit and 30-epoch patience. This is a one-seed validation experiment on CPU, not a replacement for the five-seed CUDA comparison. Python 3.11.11, PyTorch 2.7.1, NumPy 2.3.2 and four CPU threads are used on macOS 26.6.2 (arm64). Its timings are not compared with GPU records.

Each width retains the checkpoint with the best posterior-mean proxy. We then evaluate

$$\hat{\mathcal{L}}_K = \frac{1}{nK} \sum_{i=1}^n \sum_{k=1}^K \log p_{\theta}(x_i | z_i^{(k)}) - \frac{1}{n} \sum_{i=1}^n \text{KL}\{q_{\phi}(z | x_i) \| p(z)\}, \quad z_i^{(k)} \sim q_{\phi}(z | x_i). \quad (5)$$

Here $K = 32$, the evaluation seed is 20260919, and $p_{\theta}(x | z) = \mathcal{N}(g_{\theta}(z), \frac{1}{2}I)$. The Gaussian normalizing constant is included. The same 64-dimensional normal draws are truncated for smaller widths. Monte Carlo SE is the sample SD of the 32 draw-wise validation means divided by $\sqrt{32}$; it measures evaluation noise, not training-seed uncertainty. Capacity is chosen separately by validation MSE, proxy and estimated ELBO, and choices are written before test evaluation. Thus the ELBO comparison selects capacity from *proxy-selected checkpoints*; it does not reselect every epoch using ELBO.

A separate 300-epoch, seed-42 MNIST run instruments the local hard-concrete (HC) pruning implementation. Its target comes from the new ordinary width-64 anchor. We retain native-model reconstruction, constraint, gate and multiplier trajectories and compare native and dimension-transferred outputs. This checks the behaviour of that local implementation in one condition; it is not an ARM replication or a validated cross-dataset

pruning baseline. Checkpoints for these separate MNIST ordinary-VAE and HC runs are saved.

5.5. Additional Cross-Dataset Component Validation

The additional experiments use the same four datasets and fixed splits, seeds 42, 123 and 777, the largest candidate width (64, or 256 for CIFAR-10), and 300 epochs. There are 12 ARM runs and 24 ARD runs, including two prior-centering settings. They are separate from the five-seed historical comparison and the one-seed MNIST capacity study. The three-seed summaries use sample SDs; the repeated seeds and shared splits do not constitute independent dataset replications.

The ARM implementation uses paired Bernoulli gates and per-example loss differences before averaging the gate-gradient estimate. A supplied linear-loss test checks this estimator against its analytic gradient. The reconstruction constraint uses summed squared error for every dataset, including dSprites. Its threshold is the ordinary $\beta = 1$ anchor's deterministic validation MSE multiplied by 1.10 and by the number of pixels. The network loss includes a gate-masked KL sum and a constraint multiplier, and the expected gate-count penalty is applied when the moving-average constraint is satisfied. Native evaluation uses a binary mask with gate probability greater than 0.5. We report posterior-mean validation/test MSE and validation MSE averaged over 32 posterior draws, both with that mask.

This is component validation, not exact reproduction of the source algorithm [7]. The local gates start at probability 0.5 and update immediately, rather than remaining open until the first constraint hit. The local KL is a masked sum rather than the source's active-count-normalised quantity. The anchor-derived threshold and local gate-update settings also differ. A failed local run therefore does not establish a failure of the published method. Endpoint training constraint satisfaction and deterministic validation Q are different events and are tabulated separately.

The ARD implementation fixes its training weight at one and reserves the first 1800 of 18,000 training-pool examples for prior updates; the remaining 16,200 supply parameter-gradient updates. The prior-update pool is fixed, with latent samples refreshed at epochs 1, 11, ..., 291 and after training. This differs from the source's resampled pool and tuned regularisation weight [8]. We evaluate both a zero centre in the variance update, as set in the source, and a sample-centred variance-update variant; each is trained separately. Both retain zero prior mean in the Gaussian closed-form KL.

At the final model, automatic differentiation evaluates the decoder Jacobian at posterior means for the first 100 validation examples. For prior variance v_j and decoder output coordinates ℓ , the score is

$$w_j = \frac{1}{100} \sum_{i=1}^{100} \sum_{\ell=1}^p \left(\frac{\partial g_{\ell}}{\partial z_j}(\mu(x_i)) \right)^2, \quad s_j = w_j v_j. \quad (6)$$

The readout is the number of descending scores required to explain 99% of their total. Exact refers to automatic differentiation instead of finite perturbations; it does not remove sample or training uncertainty. The recorded native ARD MSE evaluates the unpruned full-width model. It is not quality at the relevance-selected count. For both methods, dimension transfer maps the count to the nearest candidate width and evaluates the cached ordinary $\beta = 1$ VAE with the same seed.

All new runs retain evaluation-mode MSE curves every five epochs. ARM uses a 2000-example training subset; the ARD training-pool subset includes its reserved prior-update examples and is identified as such. Recorded training time includes these periodic evaluations. ARM also records final native-evaluation time; ARD separately records

Jacobian time but not final native-evaluation time. These are partial timings. The new scripts do not save model checkpoints.

An evaluation-code audit limits use of the additional ELBO fields. The 32-sample option changes MSE only: reconstruction likelihood is still evaluated at the posterior mean, while KL uses an unmasked standard-normal prior, including for ARD. These fields are not Monte Carlo ELBOs for the trained native models. We therefore use the additional records for MSE, gates, relevance counts and curves; the capacity-selection ELBO evidence remains the separate MNIST experiment in Section 5.4.

6. Results

6.1. Search Decisions and Canonical Costs

The historical MNIST expense contrast is structurally determined when the smallest qualifying width lies inside the initial window, $m_0 < m_a$, and no budget interruption occurs:

$$C_{\text{window}} - C_{\text{ascending}} = C_{\text{ref}} + \sum_{m^* < m \leq m_0} C_m \geq 0. \tag{7}$$

Both procedures already evaluate the anchor; its cost cancels. At the primary setting, $\hat{d} = 11.68$ gives $m_0 = 24$. The window requests widths 2, 4, 6, 8, 10, 12, 16, 20 and 24; ascending stops at 6. The historical ledger difference is $715.4 - 182.0 = 533.4 = 205.5 + (400.7 - 72.8)$ s (Appendix B.1). This is an accounting identity for those paths, not an empirical identification of the value of ID information.

Tables 3 and 4 evaluate policies with different stopping opportunities using uniform snapshot prices. ID-local requests nine candidates on MNIST, versus ten for the old window and four for ascending. All return the same width six. Canonical cold costs are 636, 672 and 177 s, respectively; these differ from historical ledger values because the prices are now uniform. The midpoint control requests six candidates and costs 279 s. Thus shortening the path alone does not establish an ID-specific gain.

Table 3. Retrospective search replay at $\beta = 1$, five seeds. Q : target attainment; exact: agreement with the full-grid smallest qualifying width; Δm : mean excess width. Calls include the anchor. Cost uses one canonical saved time per key plus the full reference bank for ID policies, in seconds; it is not new wall-clock timing. Gaussian likelihood except Bernoulli dSprites.

Policy	Width	Q	Exact	Δm	Calls	Cold cost
<i>MNIST</i>						
Full grid Q	6.0 ± 0.0	5/5	5/5	0.0	12.0	572
Ascending Q	6.0 ± 0.0	5/5	5/5	0.0	4.0	177
Legacy window	6.0 ± 0.0	5/5	5/5	0.0	10.0	672
ID local	6.0 ± 0.0	5/5	5/5	0.0	9.0	636
Midpoint local	6.0 ± 0.0	5/5	5/5	0.0	6.0	279
ID certified	6.0 ± 0.0	5/5	5/5	0.0	10.0	672
<i>Fashion-MNIST</i>						
Full grid Q	4.0 ± 0.0	5/5	5/5	0.0	12.0	645
Ascending Q	4.0 ± 0.0	5/5	5/5	0.0	3.0	209
Legacy window	6.4 ± 3.3	5/5	3/5	2.4	4.0	457
ID local	4.0 ± 0.0	5/5	5/5	0.0	3.0	416
Midpoint local	4.0 ± 0.0	5/5	5/5	0.0	7.0	407
ID certified	4.0 ± 0.0	5/5	5/5	0.0	3.0	416

Table 4. Retrospective search replay at $\beta = 1$, five seeds. *Q*: target attainment; exact: agreement with the full-grid smallest qualifying width; Δm : mean excess width. Calls include the anchor. Cost uses one canonical saved time per key plus the full reference bank for ID policies, in seconds; it is not new wall-clock timing. Gaussian likelihood except Bernoulli dSprites.

Policy	Width	<i>Q</i>	Exact	Δm	Calls	Cold cost
<i>dSprites</i>						
Full grid <i>Q</i>	20.0 ± 4.0	5/5	5/5	0.0	12.0	2563
Ascending <i>Q</i>	20.0 ± 4.0	5/5	5/5	0.0	9.0	1797
Legacy window	64.0 ± 0.0	5/5	0/5	44.0	4.0	999
ID local	20.0 ± 4.0	5/5	5/5	0.0	9.0	2092
Midpoint local	20.0 ± 4.0	5/5	5/5	0.0	4.0	941
ID certified	20.0 ± 4.0	5/5	5/5	0.0	9.0	2092
<i>CIFAR-10</i>						
Full grid <i>Q</i>	24.0 ± 0.0	5/5	5/5	0.0	10.0	428
Ascending <i>Q</i>	24.0 ± 0.0	5/5	5/5	0.0	4.0	157
Legacy window	131.2 ± 113.9	5/5	0/5	107.2	4.0	499
ID local	24.0 ± 0.0	5/5	5/5	0.0	4.0	498
Midpoint local	24.0 ± 0.0	5/5	5/5	0.0	5.0	213
ID certified	24.0 ± 0.0	5/5	5/5	0.0	4.0	498

Local search can reduce requests: on dSprites, midpoint-local takes four rather than nine ascending requests, with canonical costs 941 versus 1797 s and the same qualifying width in every primary seed. ID is rejected there, so ID-local follows ascending and additionally charges the bank. Across all 100 dataset–weight–seed conditions, ID-local and its certified variant match the full-grid minimum in all 100; midpoint-local matches in 98, with a maximum excess width of eight. All these new-policy outputs meet the validation target and none exceeds the original soft cost limit in the replay. These are conditional observations on a previously trained grid, not guarantees for new data.

6.2. Start Width with Reliability and Fallback Held Fixed

The additional ablation changes only the accepted start to midpoint, retaining the ID gate, ascending fallback and reference-bank charge (Table 5). Both policies select exactly the same width in all 100 conditions. ID is accepted only for the 25 MNIST conditions; there, ID-local requests 8.84 candidates versus 5.84 for the gated midpoint and costs 705.2 versus 537.7 canonical cold-use seconds. Thus the ID-derived start has no observed selection benefit and incurs more work in the stratum where it is actually used. In the other 75 conditions, both policies use the same ascending fallback.

Table 5. Start-width ablation across five weights and five candidate seeds per dataset. ID accepted comprises MNIST (25 conditions); fallback comprises the other three datasets (75). ID-local and ID-gated midpoint use identical reliability checks, fallback and reference-bank charges. The bank-free midpoint is a separate policy. Exact denotes the full-grid minimum; costs are canonical cold-use seconds, not new wall-clock measurements.

Stratum	Policy	n	Exact	Calls	Cold (s)
ID accepted	ID local	25	25/25	8.84	705.2
ID accepted	ID gated midpoint	25	25/25	5.84	537.7
ID accepted	Midpoint local	25	25/25	5.84	330.2
ID accepted	Ascending Q	25	25/25	4.12	195.1
Fallback	ID local	75	75/75	5.24	995.8
Fallback	ID gated midpoint	75	75/75	5.24	995.8
Fallback	Midpoint local	75	73/75	5.11	533.0
Fallback	Ascending Q	75	75/75	5.24	714.7
All	ID local	100	100/100	6.14	923.1
All	ID gated midpoint	100	100/100	5.39	881.3
All	Midpoint local	100	98/100	5.29	482.3
All	Ascending Q	100	100/100	4.96	584.8

The two bank-free midpoint errors occur for Fashion-MNIST (seed 31337, $\beta = 0.25$: width 12 versus minimum 6) and dSprites (seed 31337, $\beta = 4$: width 16 versus minimum 8). Both are rejected-ID conditions. Its 98/100 versus ID-local's 100/100 exact agreement therefore reflects the fallback difference, not evidence that an ID-derived start improves selection. Ascending-Q also matches all 100 minima and costs less than ID-local on every condition under cold acquisition. The common anchor guarantees attainment within these stored grids by construction; exact minimality and cost are the informative comparisons.

6.3. Coefficient, Reliability and Reuse Sensitivity

At the primary MNIST setting, changing c from one to four leaves the selected width at six but changes ID-local requests from six to eleven and cold cost from 487 to 743 s. With the original reliability thresholds, the other three datasets use identical ascending fallback at every coefficient. Their invariance is therefore not evidence that the ID coefficient is robust.

At $c = 2$, relaxing $A_{\max}$ to 0.40 admits Fashion-MNIST and raises its primary request count from three to ten without improving the width. At 0.85 it also admits CIFAR-10; admitting dSprites additionally requires the permissive $R_{\max} = 0.50$, $A_{\max} = 2.00$ stress setting. Threshold choice thus changes whether ID actually participates. Tables A4 and A5 give full primary coefficient results and representative reliability contrasts; the supplement contains every factorial row.

Across all 10,000 sensitivity rows, quality is always met, but 155 outputs miss the full-grid minimum, with maximum excess width 56. All misses occur away from $\beta = 1$. They show why local stopping cannot be certified from apparent monotonicity at one setting. These rows reuse records and reference banks; 10,000 is a count of replay configurations, not independent experimental replicates. Table 6 separates cold use, warm use and one-time bank construction shared over 25 tasks.

Table 6. ID-local costs across five weights and five candidate seeds (25 tasks per dataset). Cold charges the bank each time; warm assumes an existing bank; batch allocates its training cost once across 25 tasks. All exclude unrecorded ID/evaluation overhead. Nonmono counts qualifying masks that change from pass to fail as width increases.

Dataset	Bank (s)	Cold (s)	Warm (s)	Batch (s)	Exact	Nonmono
MNIST	207.5	705.2	497.7	506.0	25/25	1/25
Fashion-MNIST	207.1	352.9	145.8	154.1	25/25	2/25
dSprites	295.8	2127.2	1831.4	1843.3	25/25	13/25
CIFAR-10	340.5	507.2	166.7	180.3	25/25	4/25

6.4. Recorded Selector Quality

Tables 7 and 8 retain ordinary-selector outcomes at $\beta = 1$, with sample SD over five training seeds. They report model widths, validation and test error and target attainment. Their timing ledgers are separated from the canonical replay prices. The inherited ARD and GECO rows have been removed from this main comparison; their raw results and subsequent checks are implementation audits in Appendix C.2.

Table 7. Dimension selection at $\beta = 1$ (Gaussian likelihoods). Values are mean $\pm$ sample SD over five candidate-training seeds. m_{final} is the width of the evaluated ordinary VAE. MSE columns are multiplied by 10^3 ; Q counts validation target attainment. Historical timing ledgers are separated in Appendix B.1; replay costs use a single canonical snapshot.

Method	m_{final}	Val. MSE	Test MSE	Q
<i>MNIST</i>				
Grid (MSE)	12.8 ± 6.4	20.98 ± 0.24	20.50 ± 0.26	5/5
FONDUE	7.0 ± 0.0	21.93 ± 0.77	21.54 ± 0.78	5/5
FONDUE + Q	8.0 ± 0.0	21.72 ± 0.50	21.33 ± 0.55	5/5
Ascending + Q	6.0 ± 0.0	22.44 ± 0.15	22.05 ± 0.17	5/5
Legacy ID window	6.0 ± 0.0	22.44 ± 0.15	22.05 ± 0.17	5/5
<i>Fashion-MNIST</i>				
Grid (MSE)	22.8 ± 15.6	18.33 ± 0.20	18.08 ± 0.19	5/5
FONDUE	6.8 ± 0.4	19.29 ± 0.16	19.01 ± 0.16	5/5
FONDUE + Q	7.6 ± 0.9	19.00 ± 0.33	18.75 ± 0.34	5/5
Ascending + Q	4.0 ± 0.0	19.38 ± 0.24	19.06 ± 0.25	5/5
Legacy ID window	6.4 ± 3.3	19.03 ± 0.61	18.73 ± 0.57	5/5

Table 8. Dimension selection at $\beta = 1$ (Bernoulli for dSprites; Gaussian for CIFAR-10). Values are mean $\pm$ sample SD over five candidate-training seeds. m_{final} is the width of the evaluated ordinary VAE. MSE columns are multiplied by 10^3 ; Q counts validation target attainment. Historical timing ledgers are separated in Appendix B.1; replay costs use a single canonical snapshot.

Method	m_{final}	Val. MSE	Test MSE	Q
<i>dSprites</i>				
Grid (MSE)	57.6 ± 14.3	1.16 ± 0.01	1.15 ± 0.01	5/5
FONDUE	4.2 ± 0.4	5.50 ± 1.31	5.42 ± 1.28	0/5
FONDUE + Q	20.0 ± 4.0	1.26 ± 0.05	1.24 ± 0.05	5/5
Ascending + Q	20.0 ± 4.0	1.26 ± 0.05	1.24 ± 0.05	5/5
Legacy ID window	64.0 ± 0.0	1.17 ± 0.02	1.16 ± 0.03	5/5
<i>CIFAR-10</i>				
Grid (MSE)	140.8 ± 53.5	13.83 ± 0.04	14.07 ± 0.04	5/5
FONDUE	14.4 ± 0.5	17.79 ± 0.30	18.10 ± 0.30	0/5
FONDUE + Q	24.0 ± 0.0	15.12 ± 0.14	15.36 ± 0.13	5/5
Ascending + Q	24.0 ± 0.0	15.12 ± 0.14	15.36 ± 0.13	5/5
Legacy ID window	131.2 ± 113.9	14.01 ± 0.13	14.25 ± 0.14	5/5

Ordinary FONDUE returns approximately four dimensions on dSprites but has validation MSE 0.00550, versus 0.00126 for ascending search. Normal termination does not imply quality attainment. FONDUE+Q includes one budget-exceeded dSprites trial that

still returned a model; that model remains in the summary. Independent outcome counts are in Appendix A.1. The legacy ID rows for Fashion-MNIST, dSprites and CIFAR-10 use coarse-MSE fallback; the high CIFAR-10 width variability belongs to that fallback. Only CIFAR-10 is a natural colour-image benchmark here.

6.5. Additional Native Pruning and Relevance Results

Table 9 compares the historical and additional raw readouts. They are similar in some datasets but are not identical across all four: the ARM dSprites mean is 54.0 rather than the historical HC count of 64. Counts alone do not measure successful compression.

Table 9. Raw dimension readouts: historical HC and unnormalized finite-step ARD (five seeds) versus new ARM and exact-Jacobian ARD (three seeds). Mean $\pm$ sample SD. Different training and readout settings prevent attributing every difference to the estimator alone. The corrected historical finite-step ARD results are retained in Table A13.

Dataset	HC ($n = 5$)	ARM ($n = 3$)	Old ARD ($n = 5$)	Exact, zero	Exact, sample
MNIST	7.2 ± 0.4	7.7 ± 0.6	6.0 ± 0.0	6.0 ± 0.0	6.0 ± 0.0
Fashion-MNIST	4.2 ± 0.4	4.7 ± 0.6	3.8 ± 0.4	4.0 ± 0.0	4.0 ± 0.0
dSprites	64.0 ± 0.0	54.0 ± 7.2	9.2 ± 1.5	9.0 ± 0.0	9.7 ± 0.6
CIFAR-10	20.0 ± 1.0	21.0 ± 2.6	14.2 ± 0.4	26.7 ± 0.6	26.3 ± 0.6

Table 10 separates native reconstruction, dimension transfer and endpoint constraint satisfaction. For MNIST and Fashion-MNIST, all three native posterior-mean reconstructions meet Q , while only two training constraints do. All three dSprites native runs fail both checks, although their transferred ordinary VAEs meet Q . Their native MSE is approximately 0.04236, and all 64 gate probabilities remain between 0.1 and 0.9 (Appendix F); thresholded counts of 48–62 therefore do not demonstrate settled sparse solutions. CIFAR-10 meets native mean-reconstruction Q in one seed and the training constraint in two. These are limitations of the local implementation.

Table 10. New ARM-based local implementation: three seeds (42, 123, 777), 300 epochs. Widths and validation MSE ($\times 10^3$) are mean $\pm$ sample SD; parentheses give external target attainment Q . Mean and sampled reconstruction both use the learned binary gate mask; sampled uses 32 posterior draws. Transfer denotes a separate ordinary VAE at the nearest grid width. Constraint is endpoint stochastic training-constraint attainment, distinct from Q .

Dataset	Gates	Native mean (Q)	Native sampled (Q)	Transfer (Q)	Constraint
MNIST	7.7 ± 0.6	21.88 ± 0.26 (3/3)	27.36 ± 0.70 (0/3)	21.63 ± 0.70 (3/3)	2/3
Fashion-MNIST	4.7 ± 0.6	19.31 ± 0.20 (3/3)	22.76 ± 0.22 (0/3)	19.28 ± 0.23 (3/3)	2/3
dSprites	54.0 ± 7.2	42.36 ± 0.00 (0/3)	42.36 ± 0.00 (0/3)	1.17 ± 0.03 (3/3)	0/3
CIFAR-10	21.0 ± 2.6	15.33 ± 0.57 (1/3)	17.16 ± 0.34 (0/3)	16.46 ± 1.17 (1/3)	2/3

The sampled native ARM MSE exceeds the deterministic anchor-derived target in every run. This comparison demonstrates the consequence of changing the reconstruction definition; it is not a selection study under a target derived from a sampled anchor. It also explains why training-constraint status cannot be inferred from mean-reconstruction Q . The exact-Jacobian ARD zero-centre readouts are 6.0, 4.0, 9.0 and 26.7 on MNIST, Fashion-MNIST, dSprites and CIFAR-10, respectively (Table 11). The sample-centred variant gives 6.0, 4.0, 9.7 and 26.3. Small differences across these three seeds support a bounded centering-sensitivity observation, not invariance to prior specification. Both settings meet full-model and transfer Q on the three natural-image datasets and fail both on dSprites. No reduced native ARD model was evaluated.

Table 11. New exact-Jacobian ARD implementations, three seeds per setting, 300 epochs. Prior-update centre is zero or the sample latent mean. Count is a relevance readout; Full is the unpruned starting-width model, not a model reduced to that count. Transfer is an ordinary VAE at the nearest grid width. MSE $\times 10^3$, mean $\pm$ sample SD; parentheses give Q counts.

Dataset	Centre	Count	Full MSE (Q)	Transfer width	Transfer MSE (Q)
MNIST	Zero	6.0 ± 0.0	23.48 ± 0.59 (3/3)	6.0 ± 0.0	22.51 ± 0.15 (3/3)
MNIST	Sample	6.0 ± 0.0	23.61 ± 0.18 (3/3)	6.0 ± 0.0	22.51 ± 0.15 (3/3)
Fashion-MNIST	Zero	4.0 ± 0.0	19.27 ± 0.19 (3/3)	4.0 ± 0.0	19.28 ± 0.23 (3/3)
Fashion-MNIST	Sample	4.0 ± 0.0	19.43 ± 0.08 (3/3)	4.0 ± 0.0	19.28 ± 0.23 (3/3)
dSprites	Zero	9.0 ± 0.0	1.33 ± 0.02 (0/3)	8.0 ± 0.0	1.88 ± 0.08 (0/3)
dSprites	Sample	9.7 ± 0.6	1.33 ± 0.06 (0/3)	9.3 ± 1.2	1.64 ± 0.30 (0/3)
CIFAR-10	Zero	26.7 ± 0.6	14.30 ± 0.13 (3/3)	24.0 ± 0.0	15.19 ± 0.12 (3/3)
CIFAR-10	Sample	26.3 ± 0.6	14.40 ± 0.24 (3/3)	24.0 ± 0.0	15.19 ± 0.12 (3/3)

The historical unnormalised CIFAR-10 ARD count was 14.2, but the previous normalisation audit had already changed it to 26.0 (Table A13). The new counts are close to that corrected readout. Because prior updates and training runs also changed, the difference from 14.2 cannot be attributed solely to exact Jacobians. Appendix F supplies all-dataset curves, gate distributions, held-out test MSE and timing components for these additional runs.

6.6. Representation and Sample Dependence of ID

Table 12 gives the complete reference-bank summary, including Fashion-MNIST. Only MNIST passes both checks. Because the same bank is reused, five identical pass/fail decisions do not constitute five independent replications of ID estimation.

Table 12. ID-reference AE widths and TwoNN estimates (mean $\pm$ SD over three reference seeds); stability R and estimator disagreement A . Limits are $R \leq 0.15$ and $A \leq 0.25$. The same reference bank is reused across candidate seeds and β .

Dataset	Width: estimate	R	A
MNIST	55: 10.93 ± 0.08 ; 111: 11.78 ± 0.12 ; 221: 12.32 ± 0.09	0.118	0.150
Fashion-MNIST	49: 9.91 ± 0.06 ; 98: 10.45 ± 0.04 ; 197: 10.83 ± 0.05	0.088	0.361
dSprites	30: 3.96 ± 0.48 ; 61: 3.91 ± 1.12 ; 122: 2.10 ± 0.02	0.474	1.816
CIFAR-10	81: 13.60 ± 0.09 ; 162: 13.44 ± 0.71 ; 325: 12.34 ± 0.50	0.094	0.834

Figure 1 separately shows the sample-size and PCA analyses. At 1000 examples the two estimators agree within the specified 0.25 relative tolerance on all four raw-input datasets; at 10,000, only MNIST does. These are raw-input checks, distinct from the AE-latent checks used by the selector. On dSprites, input TwoNN is 7.70 in this diagnostic, PCA estimates range from 2.68 to 4.32, and individual AE estimates range from 2.08 to 4.56. The combined range is about 3.7-fold; the corresponding MNIST range is about 1.7-fold. These are representation-dependent estimates, not competing measurements of a known ground truth.

6.7. Dependence on Regularisation and Quality Context

Define the reported range for method a and dataset d by

$$S_{a,d} = \max_{\beta \in \mathcal{B}} \bar{m}_{a,d}(\beta) - \min_{\beta \in \mathcal{B}} \bar{m}_{a,d}(\beta), \quad \bar{m}_{a,d}(\beta) = \frac{1}{5} \sum_{s=1}^5 m_{a,d,s}(\beta), \quad \mathcal{B} = \{0.25, 0.5, 1, 2, 4\}. \quad (8)$$

This is the range of seed means of raw returned dimensions, not the mean of within-seed ranges. For the FONDUE-VAR row all five outputs are defined at each setting, including any raw zero. It is a descriptive range on the tested ladder, not a new parameter-free robustness index.

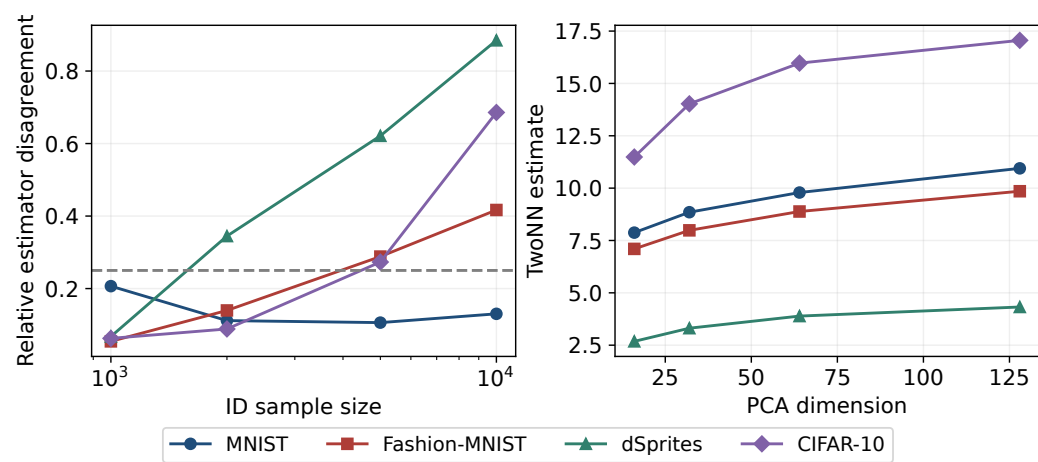


Figure 1. Left: $|d_{\text{TwoNN}} - d_{\text{MLE}}|/d_{\text{TwoNN}}$ on raw inputs as estimation sample size changes; dashed line is 0.25. Right: TwoNN after PCA preprocessing. PCA rank, sample size and representation are distinct interventions; AE results are in Table 12.

Table 13. Range of seed-mean returned dimensions across $\beta \in \{0.25, 0.5, 1, 2, 4\}$. Pruning adaptations are restricted to the implementation audit. Fixed 32 is a design constant; fallback spans are not ID robustness. Gaussian likelihood except Bernoulli dSprites.

Method	MNIST	Fashion	dSprites	CIFAR-10
Grid (MSE)	14.8	20.8	40.0	91.2
Coarse search	33.6	37.2	0.0	214.4
Fixed 32	0.0	0.0	0.0	0.0
FONDUE	3.8	0.8	0.8	8.8
FONDUE-VAR (mixed kept)	50.6	63.0	106.2	68.8
Ascending + Q	6.0	3.6	15.2	52.8
FONDUE + Q	3.6	1.6	28.0	44.8
Legacy ID window	6.0	37.2	0.0	214.4

Ranges differ by method and dataset, with clear ordering reversals. For example, ascending versus FONDUE+Q ranges are 6.0 versus 3.6 on MNIST, but 15.2 versus 28.0 on dSprites. The legacy ID window’s CIFAR-10 range of 214.4 is inherited from fallback. A fixed dimension has zero range by design; dSprites ID fallback has zero range because coarse search always returns 64. Neither demonstrates useful selection robustness. We do not average ranges across differently scaled grids or assert a transferable ranking.

The ARD call did not vary its selection-stage weight and GECCO varied an anchor-derived constraint rather than β directly. Both belong to the implementation audit, not Figure 2’s ordinary-VAE regularisation comparison.

An AU-threshold intervention holds the checkpoint fixed and recomputes Equation (4). A quality-margin intervention instead changes which cached models satisfy the target. Neither is the same as changing training β . We report these readout/quality analyses separately (Section 6.9 and Appendix C.3). No sensitivity claim is made for untested FONDUE stopping thresholds or ARD cumulative relevance fractions.

6.8. Learning Curves, Test Evaluation and Selection Criteria

Figure 3 uses actual early-stopped cache records selected by ascending-Q at the primary setting, including MNIST width six and CIFAR-10. Each run is shown only over its recorded epochs and the proxy-selected checkpoint is marked. These cache records contain validation curves but no epoch-wise train curve; no missing train measurements are imputed. The separate new MNIST run supplies matched train/validation curves for width six (Appendix E).

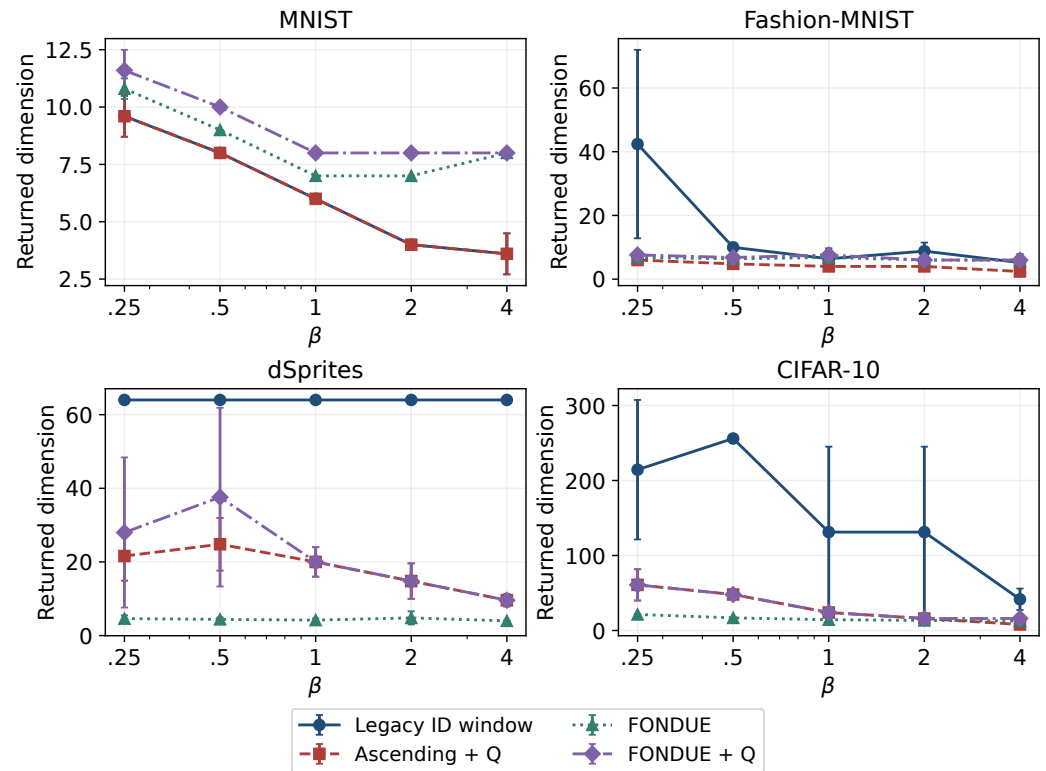


Figure 2. Raw returned dimension across the five contexts, mean $\pm$ SD over five candidate seeds. Gaussian likelihood except Bernoulli dSprites. These ordinary VAE selectors change training β . The legacy ID window falls back on Fashion-MNIST, dSprites and CIFAR-10 throughout the ladder.

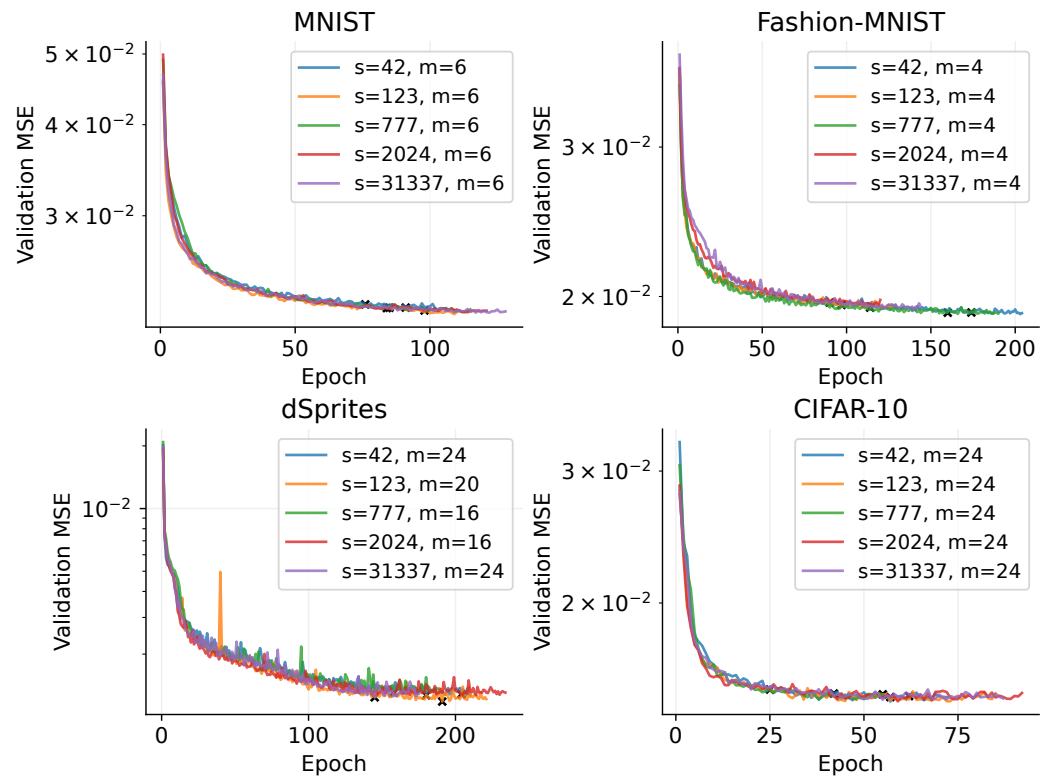


Figure 3. Actual selected-candidate validation MSE at $\beta = 1$, five seeds, Gaussian likelihood except Bernoulli dSprites. Widths are chosen by ascending-Q; black crosses mark proxy-selected epochs. Curves end at the recorded stopping time. These are not the fixed-300-epoch diagnostic runs in Figure 4.

The saved trajectory experiment supplies train/validation curves for MNIST widths 12, 32 and 64 and dSprites widths 6, 16 and 32, with five seeds each (Figure 4). Training MSE is evaluated in deterministic mode on a fixed 2000-example training subset sampled with seed 20260919. Both train and validation improve, but validation error remains higher. None of these 30 runs passes the stringent final 50-epoch plateau check; completing 300 epochs is not proof of convergence. These curves cover representative settings, not all four datasets.

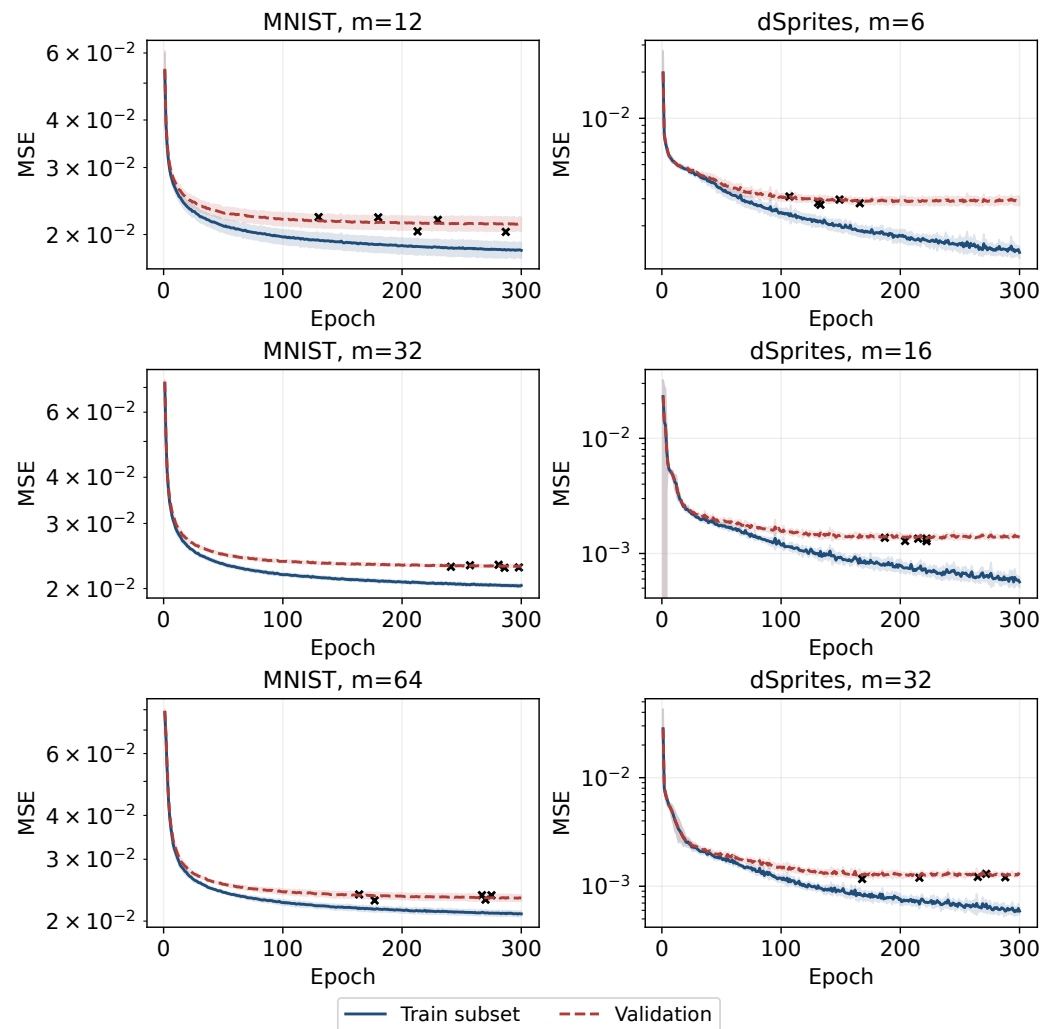


Figure 4. Train and validation MSE, means with ± 1 SD bands over five seeds. MNIST uses Gaussian and dSprites Bernoulli likelihood; $\beta = 1$. Black crosses mark each run's checkpoint selected by the validation ELBO proxy. These diagnostic runs have no early stopping.

Table 14. Selection-criterion comparison at $\beta = 1$, five seeds; test MSE $\times 10^3$ and probe test accuracy (%). Probes use class labels (shape for dSprites); only Grid (probe) uses them for capacity selection. Time is the historical ledger including anchor completion, not the canonical replay price. Recorded probe time is charged to that selector. Other displayed probe accuracies are auxiliary evaluations with uncharged costs. Likelihoods follow Table 2.

Dataset	Criterion	m	Test MSE	Accuracy	Time (s)
MNIST	Grid (MSE)	12.8 ± 6.4	20.50 ± 0.26	89.4 ± 0.9	628 ± 35
MNIST	Grid (ELBO proxy)	10.0 ± 2.0	20.54 ± 0.25	89.3 ± 0.7	628 ± 35
MNIST	Grid (probe)	51.2 ± 13.4	23.01 ± 0.55	92.7 ± 0.5	696 ± 36
Fashion-MNIST	Grid (MSE)	22.8 ± 15.6	18.08 ± 0.19	75.5 ± 2.6	645 ± 42
Fashion-MNIST	Grid (ELBO proxy)	15.6 ± 8.0	18.12 ± 0.27	74.4 ± 2.6	645 ± 42
Fashion-MNIST	Grid (probe)	57.6 ± 8.8	18.99 ± 0.46	78.4 ± 0.6	750 ± 40
dSprites	Grid (MSE)	57.6 ± 14.3	1.15 ± 0.01	47.7 ± 2.3	2565 ± 82
dSprites	Grid (ELBO proxy)	57.6 ± 14.3	1.15 ± 0.01	47.7 ± 2.3	2565 ± 82
dSprites	Grid (probe)	64.0 ± 0.0	1.16 ± 0.03	49.8 ± 3.0	2589 ± 84
CIFAR-10	Grid (MSE)	140.8 ± 53.5	14.07 ± 0.04	44.3 ± 2.6	428 ± 16
CIFAR-10	Grid (ELBO proxy)	192.0 ± 45.3	14.11 ± 0.08	46.1 ± 1.1	428 ± 16
CIFAR-10	Grid (probe)	256.0 ± 0.0	14.30 ± 0.12	47.3 ± 1.0	574 ± 15

Table 14 provides the previously omitted grid choices based on MSE, the ELBO proxy and a linear probe. Different objectives can select different widths and reconstruction errors. Test probe accuracy is an auxiliary supervised outcome; it does not measure unsupervised reconstruction quality. This comparison retains the same splits and cached candidates, with labelled validation selection explicitly stated.

The separate new MNIST comparison in Table 15 evaluates an actual expected log likelihood with the fixed Monte Carlo protocol in Section 5.4. It is not reconstructed from cached MSE and KL. Validation MSE selects width 12, the proxy selects 12, and Monte Carlo ELBO selects 12. Agreement in this run does not make the proxy an ELBO estimator. The complete grid scores, MC errors and selected epochs appear in Appendix E. One new training seed does not establish a dataset-level advantage of any criterion. The additional four-dataset pruning records do not extend this capacity-selection ELBO comparison, for the evaluation-code reasons in Section 5.5.

Table 15. New MNIST seed-42 capacity selection at $\beta = 1$, Gaussian variance $1/2$. All criteria choose among the same twelve proxy-selected checkpoints. MSE is multiplied by 10^3 ; ELBO is mean nats/example and its $\pm$ value is Monte Carlo SE ($K = 32$), not a training-seed SD. Test is evaluated after validation choices are frozen.

Criterion	Width	Val. MSE	Test MSE	MC ELBO $\pm$ MC SE
MSE	12	20.98	20.43	-478.067 ± 0.012
ELBO proxy	12	20.98	20.43	-478.067 ± 0.012
MC ELBO	12	20.98	20.43	-478.067 ± 0.012

6.9. Activity Readout: Trajectories, Thresholds and Prediction

Figure 5 and Table 16 compare the two counts on identical checkpoints. At MNIST width 64, epoch-five variable types give 9.8 ± 3.8 and AU gives 49.4 ± 8.1 ; at epoch 300 both give 6.4 ± 0.5 . This example distinguishes mean agreement from between-seed variability. The table also reports within-run changes from 50 to 300 epochs. Their magnitude depends on the condition, so no general stability superiority is inferred from the final agreement.

Table 16. Variable-type count V (active plus mixed) and AU count A , five seeds, $\beta = 1$, Gaussian MNIST and Bernoulli dSprites. Mean $\pm$ SD quantify between-seed variation. The last column gives mean absolute within-run changes from epoch 50 to 300 for V/A , not a temporal variance.

Dataset	m	V_5	A_5	V_{300}	A_{300}	Change
MNIST	12	7.2 ± 0.8	8.0 ± 1.6	6.4 ± 0.5	6.4 ± 0.5	0.0/0.0
MNIST	32	8.2 ± 1.5	20.8 ± 4.2	6.0 ± 0.0	6.0 ± 0.0	0.0/0.0
MNIST	64	9.8 ± 3.8	49.4 ± 8.1	6.4 ± 0.5	6.4 ± 0.5	0.4/2.6
dSprites	6	6.0 ± 0.0	6.0 ± 0.0	6.0 ± 0.0	6.0 ± 0.0	0.0/0.0
dSprites	16	16.0 ± 0.0	16.0 ± 0.0	14.2 ± 0.4	13.4 ± 0.5	1.8/1.8
dSprites	32	32.0 ± 0.0	31.8 ± 0.4	14.2 ± 1.1	14.4 ± 1.1	12.4/4.2

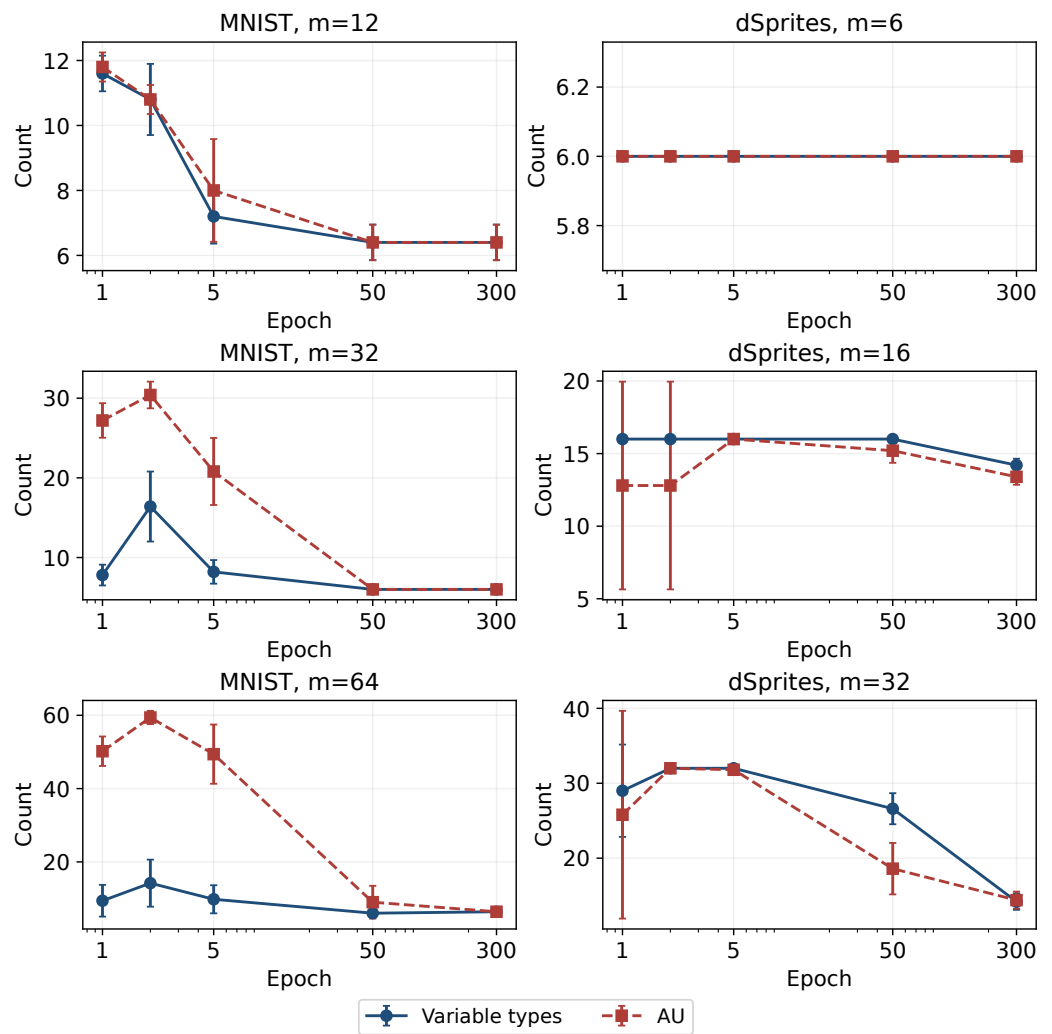


Figure 5. Variable-type count (active plus mixed) and AU from the same checkpoints at epochs 1, 2, 5, 50 and 300; means and seed SDs, with lines connecting the observed checkpoints only. $\beta = 1$, Gaussian MNIST and Bernoulli dSprites. Constant full-capacity counts are not evidence of diagnostic stability.

The threshold-only comparison in Figure 6 uses fixed $\beta = 1$ checkpoints. On CIFAR-10, $\delta = 0.001$ counts every coordinate at every tested grid width, whereas $\delta = 0.1$ counts far fewer at large widths. The inference “no unused coordinates” therefore depends on this threshold. Normalising the threshold changes the diagnostic definition; it does not establish a dimension estimate.

The predictive task asks whether the next smaller grid candidate meets the same seed’s quality target. Each row is one validation-selected checkpoint at a nonminimal grid

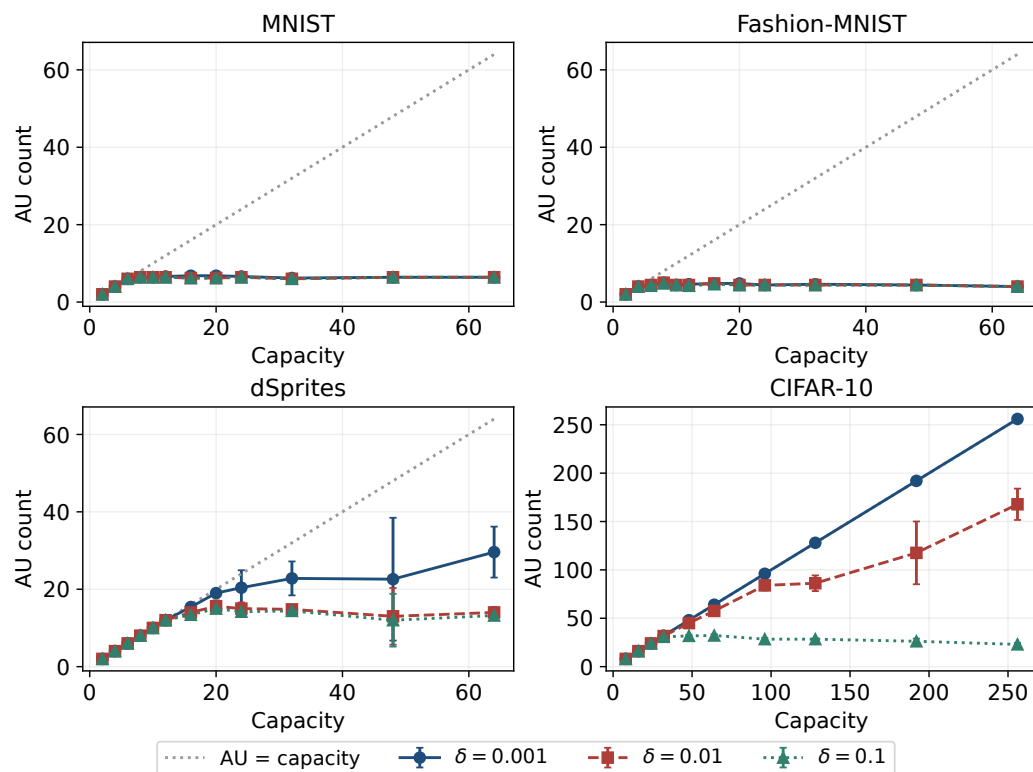


Figure 6. AU under three thresholds on the same $\beta = 1$ candidates; mean $\pm$ seed SD. Gaussian likelihood except Bernoulli dSprites. The diagonal is capacity. This is a readout-threshold intervention, unlike the training/context intervention in Figure 2.

width. Baseline features are validation MSE, KL, the ELBO proxy, width, width/input-ID ratio, and the slope of validation MSE over the last 20 recorded epochs. The three added AU features are AU, AU/width and AU/input-ID. Standardisation is fitted only on development rows. An L_2 logistic predictor is fitted on MNIST (55 rows), with $C = 10$ chosen on a three-seed/two-seed development split. Evaluation uses dSprites (55 rows), Fashion-MNIST (55) and CIFAR-10 (45), each with five seeds. The configured MNIST-CNN predictive condition was not executed.

The development split uses seeds 42, 123 and 777 for the inner fit and 2024 and 31337 for selecting C , then refits on all five. Predictors use up to 5000 iterations. Each evaluation dataset-seed block is scored over its complete set of nonminimal grid widths; checkpoint rows are not resampled as independent observations.

The original internally specified analysis reported an AU-minus-baseline AU-ROC improvement of $+0.108$, with a 2000-replicate seed-record bootstrap 95% interval $[+0.059, +0.157]$. A subsequently added control used three independent standard-normal features and one noise realisation (generator seed 7); its separately reported improvement was $+0.098$ with interval $[-0.025, +0.227]$. Overlap of these intervals does not test their difference or establish equivalence. These archived values are retained as the original analysis, not as a validated positive finding: its AUROC calculation used probabilities that can saturate numerically under the shift between datasets.

Table 17. Retrospective paired AU comparison at $\beta = 1$, using logistic decision scores to avoid saturated-probability ties. Values are means of five seed-specific AUROCs, each evaluated over all nonminimal capacities in that seed. Random features are three standard-normal columns with generator seed 7. All methods use the same rows and labels.

Evaluation condition	Rows	Base	Base + AU	Base + random	AU – random
dSprites	55	0.950	0.858	0.898	-0.041
Fashion-MNIST	55	0.980	1.000	0.880	+0.120
CIFAR-10	45	1.000	1.000	1.000	+0.000

Baseline AUROC is already 0.950, 0.980 and 1.000 on dSprites, Fashion-MNIST and CIFAR-10, respectively. In particular, CIFAR-10 has no upward AUROC headroom. The result concerns additional value on this specific, near-ceiling prediction task, not the information content of AU in general.

For this revision we reconstruct both predictors at explicitly fixed $\beta = 1$, restrict rows to the declared grid, and compute AUROC from logistic decision scores. The original cache-reading routine did not filter β or restrict the grid, so rerunning it after cache expansion would also be ambiguous. Decision scores preserve ranks that are lost when sigmoid probabilities round to exactly one. Saturation counts appear in Appendix C.3. This is a corrected retrospective analysis, not an exact replay of archived probabilities. The AU-minus-baseline difference is now -0.024 , with interval $[-0.102, +0.027]$. We also compare AU and random predictors on the same dataset-seed blocks. The paired mean difference is $+0.026$, with a retrospective hierarchical-bootstrap 95% interval $[-0.040, +0.133]$ (2000 replicates). Each replicate resamples the three evaluation datasets and then the five seed blocks within each selected dataset; all capacities and both predictors remain paired within a block. Models are held fixed, so this interval does not cover development-model fitting uncertainty. Only three evaluation datasets are available, limiting precision of across-condition inference. A secondary analysis averages the comparator over 100 noise realisations (seeds 0–99), giving a paired difference -0.007 and interval $[-0.064, +0.039]$ under the same resampling. Neither analysis establishes an AU-specific benefit in these evaluation conditions. We do not claim that AU has no information, that the predictors are equivalent, or that lack of significance proves insufficient power.

6.10. Implementation Audit and Degenerate Outputs

Two scale inconsistencies were identified in local implementations. First, the earlier VAE loss averaged KL over latent dimensions while summing reconstruction over input dimensions. Algebraically this gives $\beta_{\text{effective}} = \beta_{\text{nominal}}/m$. The image results in this paper come from the corrected sum-over-latents convention. The algebra does not imply that complete optimisation trajectories or AU counts are reproduced simply by relabelling β ; the earlier “within 1.7 units” claim is not used here.

Second, an intermediate GECO adaptation combined per-pixel-mean constraint values with other terms on incompatible scales. The corrected implementation uses summed squared error and multiplies the per-pixel target by the input dimension. Either sum or mean conventions can be valid if the objective, target and multiplier scales are consistent. The observed behaviour of this hard-concrete adaptation is not attributed to the original ARM implementation.

A further ARM-specific implementation check concerns averaging order. The supplied linear-loss test compares the per-example antithetic estimator with its analytic gradient; averaging losses before forming their correlation with the gate samples removes the required pairing. The additional training code uses per-example losses. This component correction does not imply all constraints are satisfied: endpoint counts are shown in Table 10. It is distinct from the historical loss-scaling and ARD relevance-normalisation audits.

FONDUE-VAR’s active-only setting returning zero is a separate phenomenon, not a third sum/mean defect. At MNIST’s one-epoch setting it returns zero on all five trials; no final VAE exists for these outputs. Table 18 shows both settings and their quality/validity counts.

Table 18. FONDUE-VAR at $\beta = 1$: both keep-mixed settings, valid-model and quality counts, test MSE $\times 10^3$, and recorded training time. Raw zero outputs remain in the dimension summary; nonexistent models have no MSE. Five seeds; likelihoods follow Table 2.

Dataset	Mixed kept	Raw m	Valid	Q	Test MSE	Time (s)
MNIST	yes	9.8 ± 8.5	5/5	5/5	22.53 ± 0.89	113 ± 22
MNIST	no	0.0 ± 0.0	0/5	0/5	—	63 ± 10
Fashion-MNIST	yes	14.4 ± 3.8	5/5	5/5	18.96 ± 0.39	91 ± 11
Fashion-MNIST	no	2.0 ± 0.0	5/5	0/5	25.27 ± 0.06	120 ± 18
dSprites	yes	26.0 ± 49.5	4/5	1/5	9.53 ± 9.62	412 ± 118
dSprites	no	1.2 ± 1.6	2/5	0/5	7.71 ± 0.02	278 ± 49
CIFAR-10	yes	34.2 ± 1.3	5/5	5/5	14.40 ± 0.15	103 ± 13
CIFAR-10	no	3.2 ± 0.8	5/5	0/5	31.65 ± 2.68	96 ± 13

7. Discussion

The controlled start-width ablation changes the interpretation of the ID-local result. Its 100/100 minimum-width agreement is reproduced by an ID-gated midpoint with the same fallback. Within the 25 accepted-ID conditions, the ID start requires more candidates without changing the answer. The bank-free midpoint’s two errors occur where the ID policies use ascending fallback. Consequently, this evidence does not establish an advantage of the ID-derived start. Reliability-gated search and an informative geometric starting width are distinct hypotheses.

The target includes the largest-width anchor, so quality attainment by an exhaustive upward path is guaranteed within the stored grid. Minimality and cost remain substantive outcomes. The factorial analysis finds nonminimal outputs in 155 of 10,000 settings; the default agreement is not a universal local-search guarantee. Shared reference banks, one split, retrospective policy design and four datasets limit inference. An independent reference-bank replication remains open. Acquisition and reuse costs also answer different deployment questions.

Regularisation ranges and AU-threshold responses measure different interventions. Some dimension ranges reflect fallback, grid scale or a changing quality target. A constant output may be fixed by design or by an unvaried implementation argument. The paired predictive comparison does not establish an AU-specific advantage over matched random-feature controls on its near-ceiling task. This limits the role of AU here to a diagnostic of coordinate use under a stated threshold.

The additional pruning experiments improve coverage to three seeds on all four datasets and provide native MSE, gates, relevance scores, curves and partial timing. They do not make dimension transfer equivalent to native compression. dSprites illustrates this distinction particularly clearly: failed native ARM models transfer counts to successful ordinary VAEs. ARD’s full-width quality likewise cannot certify quality after native pruning. Exact-Jacobian readouts remove finite-step error, while training and prior-update choices remain part of the implementation. The Gaussian KL approximation is shared with the ARD source; the sample-centred variance update is a sensitivity variant, not a necessary correction of the source’s zero-centre setting.

The additional four-dataset ELBO fields do not compute the expected likelihood and native prior KL needed for the claimed quantity. Valid MC ELBO capacity-selection evidence remains one MNIST seed and proxy-selected checkpoints. Likewise, new pruning curves do not recover missing historical ordinary-VAE curves. Full source-algorithm re-

production, independent-bank validation, complete end-to-end timing and clean synthetic held-out evaluation remain uncompleted. The synthetic diagnostics (Appendix D) describe scale and coordinate effects; they do not show that topology determines active-axis counts.

8. Conclusions

We evaluated intrinsic-dimension-guided VAE bottleneck selection and active-unit diagnostics under explicit quality and cost definitions. Holding reliability checks, fallback and reference costs fixed, an ID-derived start and midpoint select identical widths in all 100 conditions; the ID start requires more candidates where it is used. The apparent accuracy difference from an ungated midpoint is explained by fallback. Local-search minimality also depends on reliability and coefficient settings. Additional ARM and exact-Jacobian ARD experiments separate native quality, transferred-width quality and training constraints, and reveal remaining implementation and measurement limits. AU does not establish additional predictive value on the present near-ceiling task. These results support a bounded evaluation of ID guidance and latent diagnostics, rather than a general efficiency or pruning guarantee.

Author Contributions: Conceptualization, C.O., M.D. and K.J.; methodology, C.O. and K.J.; software, C.O. and M.D.; validation, C.O., M.D. and K.J.; formal analysis, C.O. and K.J.; investigation, C.O., M.D. and K.J.; writing—original draft preparation, C.O. and K.J.; writing—review and editing, C.O., M.D. and K.J.; supervision and project administration, K.J.

Funding: This research was funded by JSPS KAKENHI Grant Numbers JP24K15115 and JP26K14984, by the Tokyo City University Priority Research [Intelligent Technology Optimization Research Unit: ORBIT], by a Cooperative Research Project of the Research Institute of Electrical Communication (RIEC), Tohoku University, and by a Research Grant from the Telecommunications Advancement Foundation (TAF).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The image datasets are publicly available from their cited sources. The accompanying reviewer supplement MAKE52_reproducibility.zip contains configuration, historical and additional result records, analysis and training code, split indices, figures, separate MNIST validation checkpoints and a SHA-256 manifest. Extract the archive and follow its README.md; Appendix G maps its contents. Historical MAKE49 and additional MAKE52 pruning checkpoints are unavailable. The archive is supplied with this revision; it is not a public data deposit and has no DOI.

Acknowledgments: Generative-AI assistance was used for experimental-code drafting and refactoring and English-language editing in the earlier manuscript preparation. OpenAI Codex assisted with the present record audit, reaggregation, model-validation analysis, figure preparation and manuscript revision. Responsibility for the scientific content rests with the authors.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A. Quality Targets and Outcome Accounting

Table A1. Quality anchors and targets, in units of 10^{-3} MSE, mean $\pm$ SD over five seeds. Each decision uses its own seed-specific target, not the displayed mean. The relative margin is 10%; the absolute floor is inactive. Gaussian likelihood except Bernoulli dSprites.

Dataset	β	Anchor MSE	Target $T_{D,s}$
MNIST	0.25	15.843 $\pm$ 0.562	17.428 $\pm$ 0.618
MNIST	0.5	19.202 $\pm$ 0.497	21.122 $\pm$ 0.546
MNIST	1	23.712 $\pm$ 0.532	26.084 $\pm$ 0.585
MNIST	2	30.061 $\pm$ 0.560	33.068 $\pm$ 0.617
MNIST	4	38.977 $\pm$ 0.673	42.875 $\pm$ 0.740
Fashion-MNIST	0.25	15.591 $\pm$ 0.678	17.150 $\pm$ 0.746
Fashion-MNIST	0.5	17.408 $\pm$ 0.513	19.148 $\pm$ 0.564
Fashion-MNIST	1	19.679 $\pm$ 0.162	21.647 $\pm$ 0.179
Fashion-MNIST	2	22.416 $\pm$ 0.527	24.658 $\pm$ 0.580
Fashion-MNIST	4	25.967 $\pm$ 0.587	28.564 $\pm$ 0.645
dSprites	0.25	1.015 $\pm$ 0.032	1.117 $\pm$ 0.036
dSprites	0.5	1.040 $\pm$ 0.032	1.144 $\pm$ 0.035
dSprites	1	1.170 $\pm$ 0.024	1.287 $\pm$ 0.026
dSprites	2	1.453 $\pm$ 0.038	1.598 $\pm$ 0.041
dSprites	4	2.144 $\pm$ 0.064	2.358 $\pm$ 0.070
CIFAR-10	0.25	9.407 $\pm$ 0.610	10.348 $\pm$ 0.671
CIFAR-10	0.5	11.151 $\pm$ 0.200	12.266 $\pm$ 0.220
CIFAR-10	1	14.064 $\pm$ 0.108	15.470 $\pm$ 0.119
CIFAR-10	2	17.806 $\pm$ 0.262	19.587 $\pm$ 0.288
CIFAR-10	4	22.399 $\pm$ 0.286	24.638 $\pm$ 0.314

Appendix A.1. Independent Outcome Counts

The following tables separate the original combined state from flags and final-model quality. A budget flag can coexist with ID fallback. The pruning wrapper did not enforce the ordinary selector budget during training, so its zero budget-flag count is not evidence of equal budget control. All valid returned models, including those from budget-exceeded trials, contribute to the primary means. Degenerate outputs contribute to raw-count and cost summaries but not to MSE.

Table A2. Outcome audit at $\beta = 1$; all counts have denominator five. S: recorded success; I: ID unreliable; B: budget exceeded; D: degenerate; N: no compact alternative. FB is the fallback flag; budget, validation quality Q , and compression of the final VAE are independently recomputed or read from flags.

Dataset	Method	State	FB	Budget	Q	Compact
MNIST	Grid (MSE)	S5	0	0	5	5
MNIST	FONDUE	S5	0	0	5	5
MNIST	FONDUE + Q	S5	0	0	5	5
MNIST	GECO adaptation	S5	0	0	5	5
MNIST	ARD adaptation	S5	0	0	5	5
MNIST	Ascending + Q	S5	0	0	5	5
MNIST	Legacy ID window	S5	0	0	5	5
MNIST	FONDUE-VAR (mixed kept)	S5	0	0	5	5
MNIST	FONDUE-VAR (active only)	D5	0	0	0	0
Fashion-MNIST	Grid (MSE)	S5	0	0	5	5
Fashion-MNIST	FONDUE	S5	0	0	5	5
Fashion-MNIST	FONDUE + Q	S5	0	0	5	5
Fashion-MNIST	GECO adaptation	S5	0	0	5	5
Fashion-MNIST	ARD adaptation	S5	0	0	4	5
Fashion-MNIST	Ascending + Q	S5	0	0	5	5
Fashion-MNIST	Legacy ID window	I5	5	0	5	5
Fashion-MNIST	FONDUE-VAR (mixed kept)	S5	0	0	5	5
Fashion-MNIST	FONDUE-VAR (active only)	S5	0	0	0	5

Table A3. Outcome audit at $\beta = 1$; all counts have denominator five. S: recorded success; I: ID unreliable; B: budget exceeded; D: degenerate; N: no compact alternative. FB is the fallback flag; budget, validation quality Q , and compression of the final VAE are independently recomputed or read from flags.

Dataset	Method	State	FB	Budget	Q	Compact
dSprites	Grid (MSE)	S5	0	0	5	1
dSprites	FONDUE	S5	0	0	0	5
dSprites	FONDUE + Q	B1, S4	0	1	5	5
dSprites	GECO adaptation	S5	0	0	5	0
dSprites	ARD adaptation	S5	0	0	0	5
dSprites	Ascending + Q	S5	0	0	5	5
dSprites	Legacy ID window	I5	5	0	5	0
dSprites	FONDUE-VAR (mixed kept)	D1, S4	0	0	1	3
dSprites	FONDUE-VAR (active only)	D3, S2	0	0	0	2
CIFAR-10	Grid (MSE)	S5	0	0	5	5
CIFAR-10	FONDUE	S5	0	0	0	5
CIFAR-10	FONDUE + Q	S5	0	0	5	5
CIFAR-10	GECO adaptation	S5	0	0	2	5
CIFAR-10	ARD adaptation	S5	0	0	0	5
CIFAR-10	Ascending + Q	S5	0	0	5	5
CIFAR-10	Legacy ID window	I5	5	0	5	3
CIFAR-10	FONDUE-VAR (mixed kept)	S5	0	0	5	5
CIFAR-10	FONDUE-VAR (active only)	S5	0	0	0	5

Appendix B. Search Sensitivity and Timing Lineage

These tables use the retrospective policy in Algorithm 1. They do not redefine the original experiments.

Table A4. Coefficient sensitivity for ID-local replay at $\beta = 1$, with $R \leq 0.15$ and $A \leq 0.25$. All five coefficients are reported; rejected datasets use the same ascending fallback. Five seeds; likelihoods as in Table 2.

Dataset	c	Width	Q	Exact	Calls	Cold cost (s)
MNIST	1	6.0 ± 0.0	5/5	5/5	6.0	487
MNIST	1.5	6.0 ± 0.0	5/5	5/5	8.0	584
MNIST	2	6.0 ± 0.0	5/5	5/5	9.0	636
MNIST	3	6.0 ± 0.0	5/5	5/5	11.0	743
MNIST	4	6.0 ± 0.0	5/5	5/5	11.0	743
Fashion-MNIST	1	4.0 ± 0.0	5/5	5/5	3.0	416
Fashion-MNIST	1.5	4.0 ± 0.0	5/5	5/5	3.0	416
Fashion-MNIST	2	4.0 ± 0.0	5/5	5/5	3.0	416
Fashion-MNIST	3	4.0 ± 0.0	5/5	5/5	3.0	416
Fashion-MNIST	4	4.0 ± 0.0	5/5	5/5	3.0	416
dSprites	1	20.0 ± 4.0	5/5	5/5	9.0	2092
dSprites	1.5	20.0 ± 4.0	5/5	5/5	9.0	2092
dSprites	2	20.0 ± 4.0	5/5	5/5	9.0	2092
dSprites	3	20.0 ± 4.0	5/5	5/5	9.0	2092
dSprites	4	20.0 ± 4.0	5/5	5/5	9.0	2092
CIFAR-10	1	24.0 ± 0.0	5/5	5/5	4.0	498
CIFAR-10	1.5	24.0 ± 0.0	5/5	5/5	4.0	498
CIFAR-10	2	24.0 ± 0.0	5/5	5/5	4.0	498
CIFAR-10	3	24.0 ± 0.0	5/5	5/5	4.0	498
CIFAR-10	4	24.0 ± 0.0	5/5	5/5	4.0	498

Table A5. Reliability-threshold sensitivity at $\beta = 1$, $c = 2$. ID: local ID start; FB: ascending fallback. Extreme tolerances are stress conditions, not recommended thresholds. Each result uses five seeds and the same reference bank. The full factorial analysis is supplied in the supplement.

$R_{\max} / A_{\max}$	Dataset	Use	Mean width	Q	Exact	Calls
0.1/0.25	MNIST	FB	6.0	5/5	5/5	4.0
0.1/0.25	Fashion-MNIST	FB	4.0	5/5	5/5	3.0
0.1/0.25	dSprites	FB	20.0	5/5	5/5	9.0
0.1/0.25	CIFAR-10	FB	24.0	5/5	5/5	4.0
0.15/0.25	MNIST	ID	6.0	5/5	5/5	9.0
0.15/0.25	Fashion-MNIST	FB	4.0	5/5	5/5	3.0
0.15/0.25	dSprites	FB	20.0	5/5	5/5	9.0
0.15/0.25	CIFAR-10	FB	24.0	5/5	5/5	4.0
0.25/0.25	MNIST	ID	6.0	5/5	5/5	9.0
0.25/0.25	Fashion-MNIST	FB	4.0	5/5	5/5	3.0
0.25/0.25	dSprites	FB	20.0	5/5	5/5	9.0
0.25/0.25	CIFAR-10	FB	24.0	5/5	5/5	4.0
0.15/0.4	MNIST	ID	6.0	5/5	5/5	9.0
0.15/0.4	Fashion-MNIST	ID	4.0	5/5	5/5	10.0
0.15/0.4	dSprites	FB	20.0	5/5	5/5	9.0
0.15/0.4	CIFAR-10	FB	24.0	5/5	5/5	4.0
0.15/0.85	MNIST	ID	6.0	5/5	5/5	9.0
0.15/0.85	Fashion-MNIST	ID	4.0	5/5	5/5	10.0
0.15/0.85	dSprites	FB	20.0	5/5	5/5	9.0
0.15/0.85	CIFAR-10	ID	24.0	5/5	5/5	4.0
0.5/2	MNIST	ID	6.0	5/5	5/5	9.0
0.5/2	Fashion-MNIST	ID	4.0	5/5	5/5	10.0
0.5/2	dSprites	ID	20.0	5/5	5/5	6.0
0.5/2	CIFAR-10	ID	24.0	5/5	5/5	4.0

Appendix B.1. Historical and Canonical Time Accounts

Table A7 preserves the MAKE50 historical ledger, including anchor completion where the original ledger omitted it. It is not the price basis of the new replay. For example, MNIST’s mean anchor prices are 69.3 s in several early ledgers and 62.6 s in the later canonical snapshot; width-six final-model prices are 39.9 and 37.6 s, respectively. The logical conditions remain `e300|es1|b1` in these matches. Identical metric summaries and requested conditions do not reveal why two timing snapshots differ, nor prove bitwise model identity. No unsupported cause such as different hardware or stopping settings is assigned.

The supplementary `results/make51/cost_lineage.csv` gives each dataset–method–weight–seed anchor/final role, complete key, original trace key, file/record source, original duration, canonical duration and available metric-identity check. Unretained off-grid candidates are marked as unavailable. An absent historical anchor charge is distinguished from a zero duration. The replay only requests retained grid candidates, so every requested anchor and final model uses a single available canonical metric/price record.

Table A6. Anchor/final timing audit at $\beta = 1$. Entries are method-role-seed records, not independent training runs. Different prices can be attached to the same logical cache key; full key, source and both durations are supplied in `cost_lineage.csv`. Stored summary metrics do not establish checkpoint identity.

Dataset	Matched entries	Different times	Max. difference (s)
MNIST	117	107	14.53
Fashion-MNIST	130	0	0.00
dSprites	119	109	0.42
CIFAR-10	130	0	0.00

Table A7. Historical MNIST training-time decomposition at $\beta = 1$ (mean seconds, five seeds). The final model is charged once and removed from the other-candidate column. ID computation and post-training evaluation were not separately timed. FONDUE epoch calibration adds 4.14 s once per dataset- β setting and is not included below.

Method	Anchor	ID AEs	Other training	Final	Total
Grid (MSE)	69.3	0.0	502.0	56.2	627.5
FONDUE	62.6	0.0	1.4	39.9	103.9
FONDUE + Q	69.3	0.0	1.4	55.3	125.9
GECO adaptation	62.6	0.0	175.4	39.6	277.6
ARD adaptation	62.6	0.0	138.0	37.6	238.2
Ascending + Q	69.3	0.0	72.8	39.9	182.0
Legacy ID window	69.3	205.5	400.7	39.9	715.4

Appendix C. Coverage and Implementation Details

Table A8. Executed coverage. Each image method row has five training seeds at each listed context. Additional diagnostic experiments have their own seed units.

Experiment	Dataset coverage	Contexts / counts
Fourteen historical ordinary selectors/controls	All four image datasets	Five β values; $14 \times 4 \times 5 \times 5 = 1400$ method records
Historical HC adaptation	All four	Five anchor-derived constraints; 100 records
Historical ARD adaptation	All four	Fixed selection-stage weight one; five repeated contexts, 100 records
ID-reference bank	All four	Three widths $\times$ three seeds each, shared across contexts
Train/validation and readout trajectories	MNIST: 12, 32, 64; dSprites: 6, 16, 32	$\beta = 1$, five seeds each; 30 runs
AU prediction	Develop: MNIST; evaluate: dSprites, Fashion-MNIST, CIFAR-10	$\beta = 1$; 55 development and 155 evaluation rows
Search replay and start-width ablation	All four	Seven policies, 700 rows; 10,000 coefficient/threshold settings
Separate MC ELBO capacity study	MNIST	One seed, 12 ordinary VAE widths
Additional ARM validation	All four	Three seeds, 12 runs; $\beta = 1$ anchor
Additional exact-Jacobian ARD	All four	Three seeds, two variance-update centres; 24 runs
Synthetic diagnostic	Swiss roll and torus	Six capacities, three seeds, two scaling options; 72 runs

The fourteen ordinary method records comprise MSE/ELBO-proxy/probe grid selection, coarse search, fixed width, FONDUE, two FONDUE-VAR settings, ID guidance with and without AU recording, ascending+Q, and coarse, FONDUE and FONDUE-VAR with added Q. Removing AU recording does not change the ID selection decisions; its overhead was not separately timed. Machine-readable results include all fourteen, even when only the principal comparisons are printed.

Table A9. Split identities reconstructed from the saved loader and fixed seeds. Tokens are the first 16 hexadecimal characters of SHA-256 over concatenated int64 train/validation/test index arrays, matching the loader convention. Official training/test pools have separate index spaces.

Dataset	Reconstructed split token
MNIST	cffaecf5bccbb333
Fashion-MNIST	cffaecf5bccbb333
dSprites	1ea516804f7f9ac7
CIFAR-10	55e0bcf8f4474c19

The split index arrays are reconstructed from the loader specification, not recovered as an independently archived execution-time split file. The MNIST and Fashion-MNIST tokens agree because their official pool sizes and index rules agree; their images are different. A compressed copy is supplied as `results/make51/reconstructed_split_indices.npz`. TwoNN uses Euclidean nearest-neighbour distances, excludes zero first distances and the empirical-CDF endpoint, and fits the log survival curve against the log distance ratio through the origin. The MLE cross-check averages local reciprocal mean-log-distance-ratio estimates over valid points at $k = 10$. Both estimates are floored at one in the local implementation.

Table A10. FONDUE short-training epochs and calibration time (once per dataset–context, separate from selector time). dSprites uses the source choice without calibration. A stable flag for the remaining datasets means consecutive calibration predictions agreed.

Dataset	β	Epochs	Calibration (s)	Stable
MNIST	0.25	2	9.64	yes
MNIST	0.5	2	9.55	yes
MNIST	1	1	4.14	yes
MNIST	2	2	8.12	yes
MNIST	4	2	9.38	yes
Fashion-MNIST	0.25	1	4.16	yes
Fashion-MNIST	0.5	1	3.64	yes
Fashion-MNIST	1	2	9.07	yes
Fashion-MNIST	2	3	13.70	yes
Fashion-MNIST	4	1	3.80	yes
dSprites	0.25	1	0.00	N/A (source)
dSprites	0.5	1	0.00	N/A (source)
dSprites	1	1	0.00	N/A (source)
dSprites	2	1	0.00	N/A (source)
dSprites	4	1	0.00	N/A (source)
CIFAR-10	0.25	3	30.34	yes
CIFAR-10	0.5	2	15.56	yes
CIFAR-10	1	2	22.72	yes
CIFAR-10	2	3	29.69	yes
CIFAR-10	4	4	42.98	yes

Appendix C.1. Historical Fixed-Window Procedure

The earlier procedure evaluated every candidate below its ID-derived upper width. Because its evaluated set included the quality anchor, the proposed expansion branch could not be entered. We retain this as a fixed-window audit, not an effective expanding search. Earlier V1/V2 activity checks are not selection branches; AU recording changes no selected width.

Algorithm A1 (Historical fixed-window ID procedure (audit)).

Input: splits, $\mathcal{M}$, seed s , β , 10% target margin, floor coefficient 0.001, input TwoNN d_0 , reference seeds $\{42, 123, 777\}$, thresholds $(0.15, 0.25)$, and budgets in Table 2.

1. Obtain the quality anchor and fix $T_{D,s}$ from Equation (1). In a fallback run this is an external audit step, not a constraint used by the coarse selector; its training is charged.
2. For $k \in \{4, 8, 16\}$ set $r_k = \text{round}(kd_0)$ (nearest integer; ties to even, no grid snapping). Train an FC-AE at each width for each reference seed, and compute TwoNN and MLE on the fixed 10,000 training examples. Compute R , A and $\hat{d}$.
3. If either check fails, flag ID-unreliable and evaluate the coarse subset. Its zero-based indices are the unique values of $\text{round}(\text{geomspace}(1, |\mathcal{M}|, 4)) - 1$. Return its minimum-validation-MSE candidate and continue to step 6.
4. Otherwise set $m_0 = \min\{m \in \mathcal{M} : m \geq 2\hat{d}\}$, or m_a if the set is empty. Initialise the evaluated set with the anchor. Evaluate all remaining $m \leq m_0$ in ascending order, stopping after a completed call that exceeds the budget.
5. Return the smallest target-satisfying width in the evaluated set. If it is the anchor, record no compact alternative. The anchor is always acceptable; the code’s planned expansion conditional on no acceptable evaluated candidate is consequently never reached.
6. Obtain or reuse the ordinary VAE at the returned width. For $m < 1$ return a degenerate output with no model. Retain budget flags even when a model is returned. Record AU as a diagnostic without changing m .
7. Return raw and final widths, model metrics, ID diagnostics, fallback and budget flags, quality/compression indicators and the time account.

Appendix C.2. Historical Pruning and Relevance Adaptations

The GECO adaptation uses Adam at 10^{-3} , batches of 128 in stored order, 300 epochs and hard-concrete parameters $\gamma = -0.1$, $\zeta = 1.1$ and temperature $2/3$. Initial gate log-odds are 2. The constraint is summed squared reconstruction error minus $D_{\text{input}}T_{D,s}$, with moving-average coefficient 0.99. A squared softplus multiplier is clamped to $[10^{-6}, 10^6]$ and updated by ascent on that moving-average violation. KL is gate-masked; the expected gate penalty is enabled only when the moving-average constraint is satisfied. The final readout counts deterministic gates above 0.5. Its training constraint uses squared error even on dSprites; the transferred ordinary VAE there uses Bernoulli reconstruction. Consequently the candidate likelihood convention does not imply identical pruning-stage objectives.

Table A11. Raw retained counts versus transferred ordinary-VAE widths at the primary context. Constraint denotes the GECO training moving average at the final update; Q denotes final ordinary-VAE validation quality. They evaluate different models and quantities. ARD has no GECO constraint.

Dataset	Adaptation	Raw count	Final width	Constraint	Q
MNIST	GECO	7.2 ± 0.4	6.4 ± 0.9	0/5	5/5
MNIST	ARD	6.0 ± 0.0	6.0 ± 0.0	—	5/5
Fashion-MNIST	GECO	4.2 ± 0.4	4.0 ± 0.0	1/5	5/5
Fashion-MNIST	ARD	3.8 ± 0.4	3.6 ± 0.9	—	4/5
dSprites	GECO	64.0 ± 0.0	64.0 ± 0.0	0/5	5/5
dSprites	ARD	9.2 ± 1.5	8.4 ± 1.7	—	0/5
CIFAR-10	GECO	20.0 ± 1.0	19.2 ± 4.4	5/5	2/5
CIFAR-10	ARD	14.2 ± 0.4	16.0 ± 0.0	—	0/5

For the archived GECO rows, the constraint uses sampled posterior latents and stochastic hard-concrete gates on training minibatches; it is an exponential moving average of per-example summed squared error minus the deterministic validation-anchor target multiplied by input dimension. Native validation reconstruction and trajectories were not retained. Thus 0/5 MNIST, 1/5 Fashion-MNIST, 0/5 dSprites and 5/5 CIFAR-10 endpoint constraint attainment cannot alone identify a bug or validate the implementation.

The new MNIST diagnostic retains those missing trajectories (Figure A1). Evaluation uses the full validation/test splits, posterior means or 32 posterior draws, and either deterministic continuous gate values or an explicitly binary mask. These evaluation quantities differ from stochastic training-gate reconstruction. The final training moving-average constraint is +0.045 summed-error units, with 7 counted gates and multiplier 0.844. The transferred grid width is 6. The endpoint training constraint remains violated. This diagnostic therefore does not promote the HC implementation to a validated pruning baseline.

Table A12. New MNIST HC diagnostic and dimension transfer, seed 42. $\text{MSE} \times 10^3$; target is 25.223 in the same units. HC width is the count above the gate threshold, although continuous inference retains nonbinary gate values. Sampled evaluation uses 32 latent draws and deterministic continuous gates. The ordinary VAE uses the nearest grid width.

Output / reconstruction	Width	Val. MSE	Test MSE	Q
HC, continuous/mean	7	22.424	21.950	yes
HC, continuous/sampled	7	27.019	26.519	no
HC, binary/mean	7	22.424	21.950	yes
Transferred ordinary VAE	6	22.343	21.933	yes

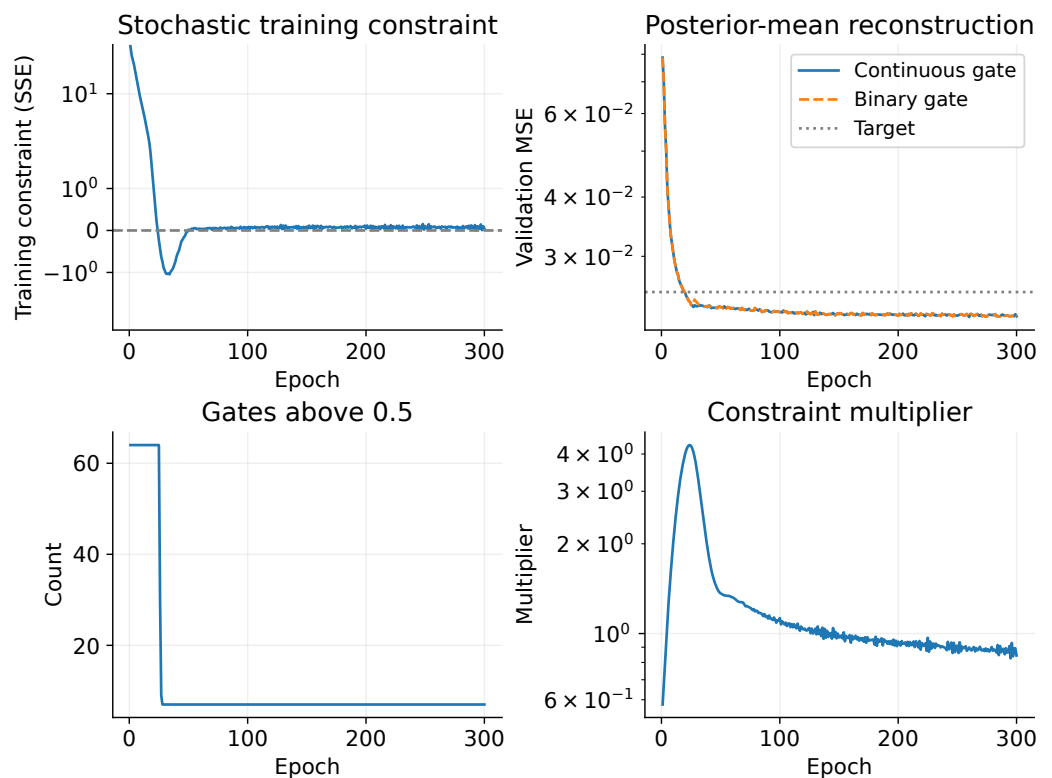


Figure A1. New MNIST hard-concrete implementation diagnostic, seed 42, 300 epochs. The training constraint uses sampled latent variables and gates; validation uses posterior means. Gate count and multiplier are shown alongside constraint and reconstruction trajectories. This is one local HC run, not an original ARM reproduction.

The ARD adaptation reserves the first 10% (1800 examples) of its training array for prior estimation and trains on the remainder. It updates $p_j = \text{mean}(\mu_j^2 + \sigma_j^2)$ every ten epochs and after training, clipping to $[10^{-6}, 10^6]$. This uses analytic posterior second moments in a fixed subset, rather than a periodically resampled encoded-data pool. Selection

training uses reconstruction plus Gaussian KL at unit weight. On the first 512 training examples it computes

$$w_j = \text{mean}_{x,k} [g_k(\mu(x) + \sqrt{p_j}e_j) - g_k(\mu(x))]^2, \quad q_j = w_j p_j, \quad (\text{A1})$$

and keeps the smallest number of descending scores reaching 99% of their total. The finite difference is not divided by its squared step. Locally it already contains a factor p_j , so multiplication by p_j again differs from a Jacobian-squared times variance score. The stored counts are therefore labelled an *ARD adaptation*; this difference cannot be removed by correcting the prose. The Gaussian approximation alone is not an original-source deviation. Identical relevance arrays and counts across contexts are checked in the numerical audit; they represent repeated fixed settings, not a β sweep.

The stored q_j and p_j are sufficient to undo the extra finite-step factor without access to the checkpoint. With $h_j = \sqrt{p_j}$, the correctly normalized finite-step score is

$$\tilde{q}_j = p_j \text{mean}_{x,k} \left[\frac{g_k(\mu + h_j e_j) - g_k(\mu)}{h_j} \right]^2 = \frac{q_j}{p_j}. \quad (\text{A2})$$

All saved p_j are positive. We recompute the cumulative 99% count and transfer it using the same nearest-grid rule. Table A13 shows that CIFAR-10 changes from a mean raw count 14.2 to 26.0, mapping to width 24 and attaining Q in all five trials. This is an algebraic correction of the finite-step readout up to stored numerical precision. Its possibly large steps and altered prior updates remain; it is not an exact-Jacobian or original-ARD benchmark.

Table A13. Corrected finite-step relevance readout from saved scores and prior variances at the primary context. The same stored perturbations are normalized by their squared step before variance weighting; no new ARD training is involved. This validates an algebraic readout correction, not exact Jacobian relevance or the source training procedure. Test MSE $\times 10^3$ is for transferred ordinary VAEs.

Dataset	Old raw	Corrected raw	Grid width	Q	Test MSE
MNIST	6.0 ± 0.0	6.2 ± 0.4	6.0 ± 0.0	5/5	22.05 ± 0.17
Fashion-MNIST	3.8 ± 0.4	4.0 ± 0.0	4.0 ± 0.0	5/5	19.06 ± 0.25
dSprites	9.2 ± 1.5	10.6 ± 0.5	10.0 ± 0.0	0/5	1.47 ± 0.05
CIFAR-10	14.2 ± 0.4	26.0 ± 0.0	24.0 ± 0.0	5/5	15.36 ± 0.13

The additional exact-Jacobian runs are reported separately in Section 6.5; they do not replace these historical records.

The FONDUE description-based classifier was checked against the public repository https://github.com/bonhemi/VAE_learning_dynamics at commit f6d6e66b536bcbe635392d3438338949d4dc8730. The local implementation uses the stated posterior-variance categories with one exponentiation and TwoNN; equivalence to the public execution path and the original experiment configuration is not asserted. As a historical reference, the source reports dSprites selection 12.2 ± 0.4 at one epoch [4], versus 4.2 ± 0.4 here at $\beta = 1$. This unmatched-setting difference is not proof of an implementation error, nor a reproduction of the original benchmark. A separate control-flow check replays the source algorithm's lower/upper-bound updates on the archived ID-gap sequence: all 100 FONDUE paths and returned widths agree with that pseudocode and terminate normally. This verifies the search update logic on those inputs; it does not validate the estimator implementation, training framework parity or the published benchmark. Its row-level results are included as `results/make51/fondue_control_flow.json`.

Appendix C.3. Threshold-Only and Quality-Margin Analyses

The retrospective predictive analysis uses Python 3.11.11, NumPy 2.3.2 and scikit-learn 1.7.1. Table A14 documents probability saturation in this reconstruction. The original predictor's complete environment and row-level probabilities were not saved, so its exact floating-point results cannot be verified from the summary alone. Row-level decision scores and probabilities for the current reconstruction are saved in `results/make51/au_predictions.csv`.

Table A14. Numerical probability saturation in the current predictor reconstruction. Entries count predictions rounded to exactly one, before AUROC calculation. Decision scores preserve ordering among these rows.

Evaluation condition	Base	Base + AU	Base + random
dSprites	55/55	44/55	49/55
Fashion-MNIST	0/55	0/55	0/55
CIFAR-10	45/45	40/45	36/45

The main AU-threshold figure uses identical checkpoints for all three δ values. Across the full ladder, the fractions of CIFAR-10 grid/seed candidates with AU equal to capacity are given below. These counts use ten widths and five seeds per β ; their rows are not 50 independent datasets.

Table A15. CIFAR-10 (Gaussian): percentage of the 50 width/seed candidates with AU equal to capacity. Thresholds are re-read on the same checkpoints within each column.

δ	$\beta = .25$	.5	1	2	4
0.001	100	100	100	98	60
0.01	60	50	36	22	10
0.1	48	38	32	20	10

For a separate retrospective quality-margin check, the complete cached grid at $\beta = 1$ is held fixed and the smallest satisfying width is read out for each margin, without retraining or changing the anchor. Table A16 reports this full-grid criterion, not the path or cost of any restricted-window selector. No FONDUE-threshold or ARD-99% sensitivity training was executed, so no response of those rules to their own readout thresholds is claimed.

Table A16. Smallest quality-satisfying cached width at $\beta = 1$ as relative margin changes; five seed means $\pm$ SD. The absolute floor 0.001 times validation variance remains fixed. Gaussian likelihood except Bernoulli dSprites.

Dataset	1%	5%	10%	25%	50%
MNIST	6.0 ± 0.0	6.0 ± 0.0	6.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0
Fashion-MNIST	4.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0	2.0 ± 0.0
dSprites	51.2 ± 17.5	36.0 ± 16.5	20.0 ± 4.0	12.0 ± 0.0	9.6 ± 0.9
CIFAR-10	41.6 ± 8.8	32.0 ± 0.0	24.0 ± 0.0	16.0 ± 0.0	16.0 ± 0.0

Appendix D. Synthetic Scale and Ablation Diagnostic

Swiss-roll points are $(t \cos t, h, t \sin t)$ with $t \sim U[1.5\pi, 4.5\pi]$ and $h \sim U[0, 1]$. Torus points are $((3 + \cos \phi) \cos \theta, (3 + \cos \phi) \sin \theta, \sin \phi)$ with independent angles uniform on $[0, 2\pi]$. Each has 5000 examples, generation seed zero, and independent Gaussian coordinate noise with SD 0.01. The first 4500 are used for fitting and the remaining 500 for the reported diagnostic. Per-coordinate standardisation in the stored experiment was fitted to all 5000 before splitting. This is disclosed as preprocessing leakage; the results are descriptive, not clean held-out test estimates.

The FC VAE uses widths 128 and 64, reversed in its decoder, ReLU hidden activations, a linear output, Adam at 10^{-3} , batch 128, 300 epochs, $\beta = 1$ and seeds 42, 123 and 777. At the last checkpoint, coordinates with posterior-mean variance at most 0.01 on the 500 diagnostic examples are zeroed and MSE is recomputed on those same examples. Table A17 reports the standardized runs. There is no separate test set in this stored experiment.

Table A17. Synthetic scale/ablation diagnostic, Gaussian likelihood, $\beta = 1$, $\delta = 0.01$, three training seeds. Data were standardized before the split; these are descriptive validation results, not leakage-free generalization estimates. Changes are mean per-run relative MSE changes (%).

Data	m	AU	MSE before	MSE after	Change
Swiss roll	1	1.0 ± 0.0	0.3511 ± 0.0267	0.3511 ± 0.0267	+0.000
Swiss roll	2	2.0 ± 0.0	0.1997 ± 0.0177	0.1997 ± 0.0177	+0.000
Swiss roll	3	3.0 ± 0.0	0.1407 ± 0.0018	0.1407 ± 0.0018	+0.000
Swiss roll	4	3.0 ± 0.0	0.1404 ± 0.0136	0.1406 ± 0.0135	+0.147
Swiss roll	6	3.0 ± 0.0	0.1365 ± 0.0023	0.1368 ± 0.0022	+0.183
Swiss roll	8	3.0 ± 0.0	0.1437 ± 0.0089	0.1440 ± 0.0089	+0.165
Torus	1	1.0 ± 0.0	0.3217 ± 0.0163	0.3217 ± 0.0163	+0.000
Torus	2	2.0 ± 0.0	0.1654 ± 0.0242	0.1654 ± 0.0242	+0.000
Torus	3	3.0 ± 0.0	0.1083 ± 0.0036	0.1083 ± 0.0036	+0.000
Torus	4	3.0 ± 0.0	0.1124 ± 0.0050	0.1124 ± 0.0050	+0.063
Torus	6	3.0 ± 0.0	0.1095 ± 0.0035	0.1096 ± 0.0035	+0.107
Torus	8	3.0 ± 0.0	0.1104 ± 0.0069	0.1105 ± 0.0069	+0.156

Both noise-free generating surfaces have intrinsic dimension two and are represented here in three ambient coordinates. A Swiss-roll patch does not thereby have minimum embedding dimension three. Observing three active coordinates under this loss is not a topological theorem. The earlier manuscript said its synthetic coordinates were standardized; the implementation audit found a mismatch between that description and the executed preprocessing. On the unstandardized Swiss roll the height coordinate has much smaller variance than the other coordinates, making its omission inexpensive in squared error. This scale observation does not alone prove the complete causal explanation of every earlier AU result.

Appendix E. New MNIST Model Records

The new experiment uses all twelve declared widths, training seed 42, and the original split reconstructed from its fixed seeds. The model architecture and objective match the ordinary MNIST protocol, while the CPU environment differs from the archived CUDA runs. Table A18 gives every proxy-selected checkpoint's validation criteria. MC SE refers only to evaluation noise over posterior draws. The frozen capacity-choice file precedes final test evaluation. Weights, epoch curves and draw-wise ELBO means are included in the reviewer supplement. No between-training-seed confidence interval is inferred from this one-seed validation.

Table A18. Complete newly trained MNIST grid, seed 42, $\beta = 1$, Gaussian likelihood. Epoch selected by the deterministic proxy; stop is the actual final training epoch. Proxy omits the Gaussian constant and posterior expectation; MC ELBO includes both. $\text{MSE} \times 10^3$.

Width	Epoch	Stop	Val. MSE	Proxy	MC ELBO $\pm$ SE
2	81	111	37.406	-34.647	-484.386 ± 0.009
4	85	115	27.840	-29.304	-480.108 ± 0.009
6	169	199	22.343	-26.696	-478.174 ± 0.010
8	73	103	22.417	-26.544	-478.285 ± 0.009
10	70	100	22.954	-26.663	-478.627 ± 0.009
12	196	226	20.983	-26.055	-478.067 ± 0.012
16	107	137	23.030	-26.774	-478.592 ± 0.009
20	144	174	21.327	-26.156	-478.134 ± 0.010
24	132	162	22.977	-26.956	-478.664 ± 0.008
32	136	166	23.402	-27.042	-478.869 ± 0.012
48	114	144	24.177	-27.411	-479.310 ± 0.013
64	217	247	22.930	-26.946	-479.043 ± 0.011

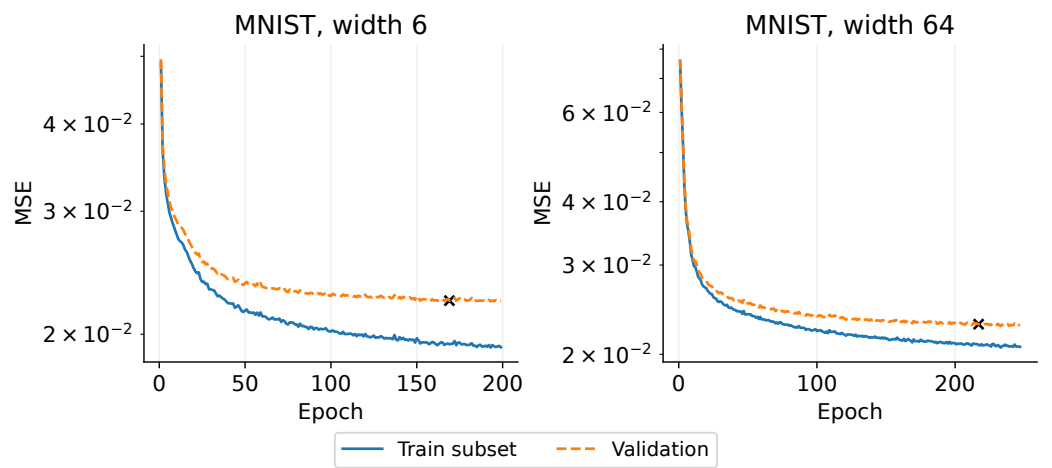


Figure A2. New MNIST seed-42 train/validation MSE at widths six and 64, Gaussian likelihood and $\beta = 1$. Training curves use a fixed 2000-example subset; validation uses all 2000 held-out examples. Black crosses mark proxy-selected epochs. These are new saved runs, not reconstructed train curves for the historical CUDA checkpoints.

Appendix F. Additional GPU Validation Records

The new records comprise three seeds per dataset for ARM and for each ARD variance-update setting. Gate bins in Table A19 and the individual constraint trajectories in Figure A3 show why a thresholded dimension count is insufficient to diagnose successful pruning. In particular, dSprites does not approach a decisive binary gate configuration or a satisfied constraint.

Table A19. ARM gate probabilities after 300 epochs. Mid denotes $0.1 \leq p \leq 0.9$; the other bins use strict inequalities. Expected is $\sum_j p_j$; count uses $p_j > 0.5$. No dSprites gate reaches either extreme, so its threshold count should not be read as a stable binary architecture.

Dataset	Seed	$p < 0.1$	Mid	$p > 0.9$	Expected	Count
MNIST	42	56	0	8	8.33	8
MNIST	123	56	1	7	8.11	8
MNIST	777	57	0	7	7.34	7
Fashion-MNIST	42	59	0	5	5.32	5
Fashion-MNIST	123	59	0	5	5.30	5
Fashion-MNIST	777	60	0	4	4.35	4
dSprites	42	0	64	0	32.08	52
dSprites	123	0	64	0	32.20	62
dSprites	777	0	64	0	32.05	48
CIFAR-10	42	236	1	19	25.65	19
CIFAR-10	123	230	3	23	30.27	24
CIFAR-10	777	235	1	20	26.18	20

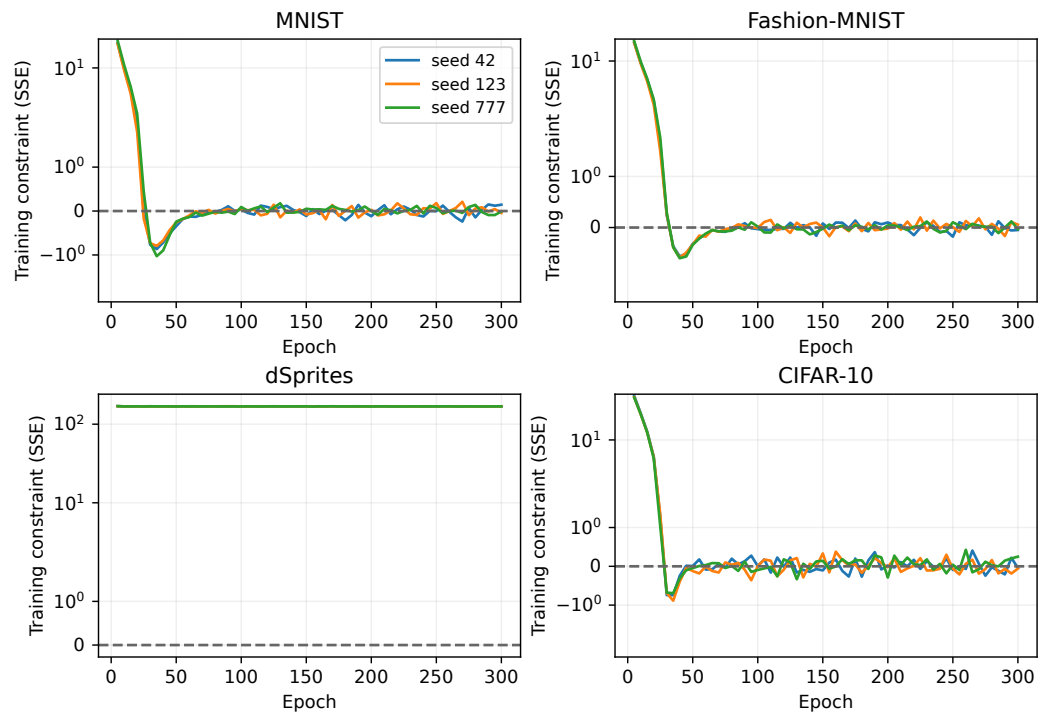


Figure A3. Moving-average stochastic training constraint for each new ARM seed, recorded every five epochs. Zero is the constraint boundary; negative values satisfy it. A symmetric logarithmic vertical scale accommodates large positive violations and negative values. These trajectories are distinct from deterministic validation-target attainment.

Figures A4 and A5 report evaluation-mode reconstruction curves for all four datasets. They describe the supplied 300-epoch runs; a visually flat curve does not certify an optimum. The ARD figure uses zero-centre variance updates. All 24 ARD curve records, including the sample-centred variant, are in the supplement.

Table A20 reports held-out test reconstruction at the fixed final native models and transferred ordinary widths. Selection uses the constraint or validation-based relevance readout, not these test metrics. The full-model ARD column is deliberately distinguished from quality at a selected native dimension.

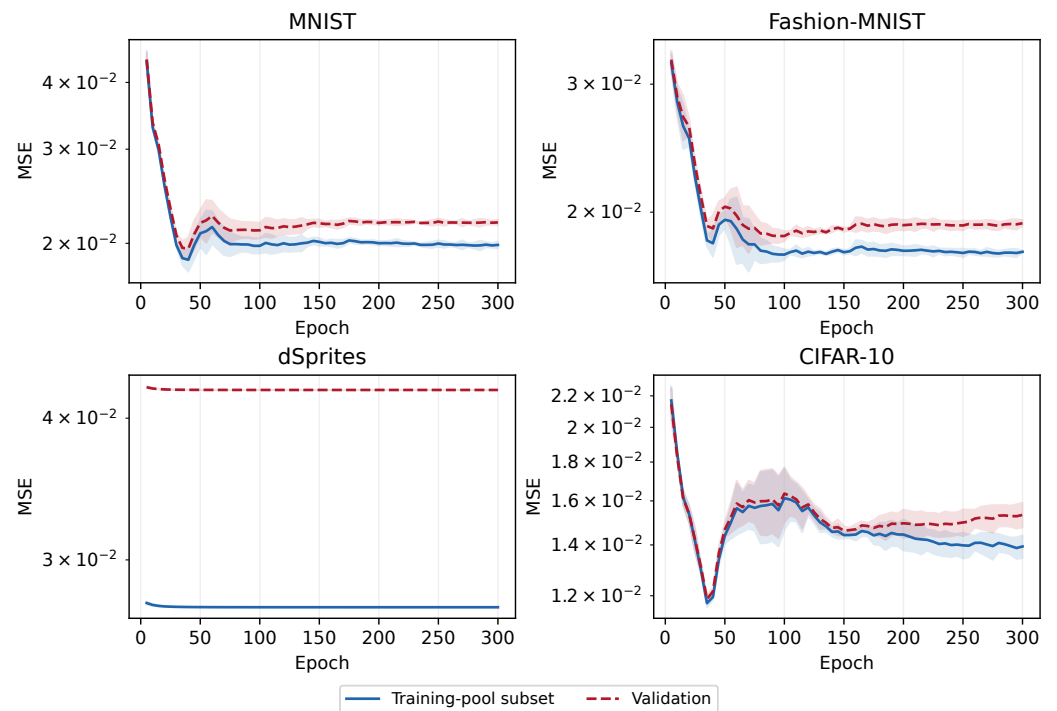


Figure A4. Native ARM posterior-mean MSE with the current binary gate mask, evaluated every five epochs on a 2000-example training subset and the validation split. Lines and shading are mean and sample SD over three seeds. Vertical axes are logarithmic; a shaded lower bound is clipped above zero for display. These curves do not reconstruct missing historical ordinary-VAE training records.

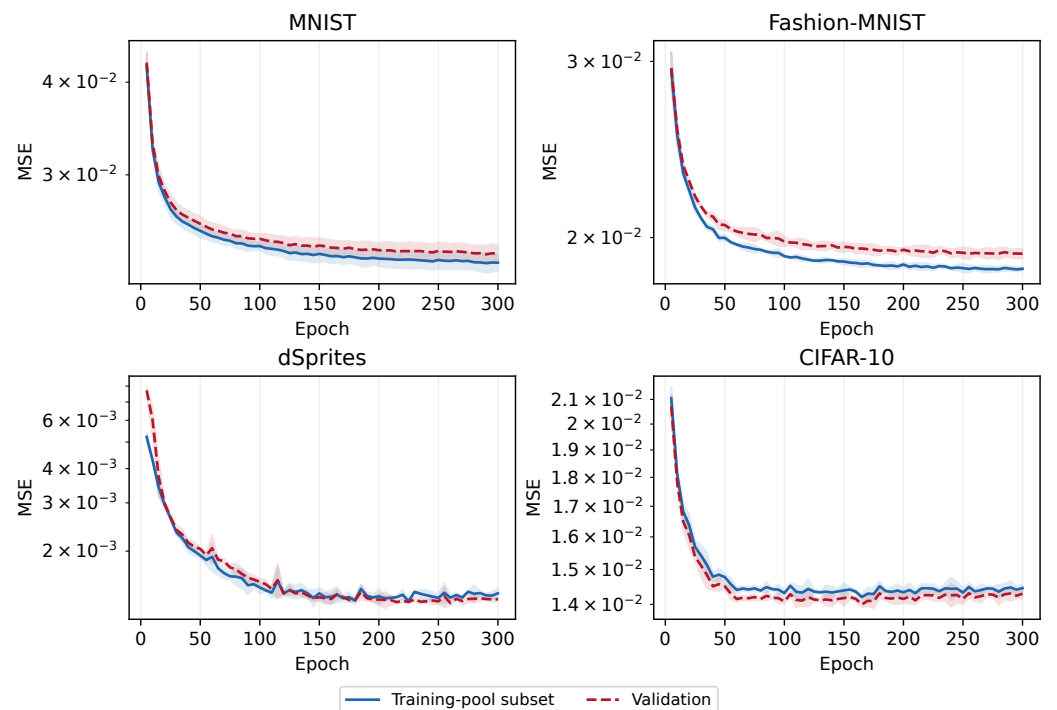


Figure A5. Full-width ARD posterior-mean MSE with zero-centre variance updates, evaluated every five epochs. Mean and sample SD over three seeds; logarithmic vertical axes. The 2000-example training-pool subset includes 1800 examples reserved for prior updates and 200 used in parameter-gradient training, so its curve is not a pure training-fit estimate. No native pruning is applied in these evaluations.

Table A20. Test MSE ($\times 10^3$), mean $\pm$ sample SD over three seeds. ARM native evaluation uses a binary gate mask and posterior mean. ARD native evaluation is the unpruned full-width model. Transferred ordinary VAEs are reused from the archived candidate cache; their test summaries did not select the new raw counts.

Dataset	Implementation	Native/full	Transferred
MNIST	ARM	21.35 $\pm$ 0.25	21.22 $\pm$ 0.79
MNIST	ARD zero	22.97 $\pm$ 0.53	22.15 $\pm$ 0.14
MNIST	ARD sample	23.17 $\pm$ 0.17	22.15 $\pm$ 0.14
Fashion-MNIST	ARM	19.01 $\pm$ 0.23	18.94 $\pm$ 0.23
Fashion-MNIST	ARD zero	18.97 $\pm$ 0.17	18.94 $\pm$ 0.23
Fashion-MNIST	ARD sample	19.17 $\pm$ 0.15	18.94 $\pm$ 0.23
dSprites	ARM	42.10 $\pm$ 0.00	1.17 $\pm$ 0.04
dSprites	ARD zero	1.32 $\pm$ 0.01	1.86 $\pm$ 0.09
dSprites	ARD sample	1.32 $\pm$ 0.06	1.61 $\pm$ 0.30
CIFAR-10	ARM	15.60 $\pm$ 0.59	16.74 $\pm$ 1.21
CIFAR-10	ARD zero	14.55 $\pm$ 0.13	15.42 $\pm$ 0.13
CIFAR-10	ARD sample	14.64 $\pm$ 0.24	15.42 $\pm$ 0.13

Table A21 reports logged duration components. The source records identify an NVIDIA GPU and the software environment. Training intervals include periodic evaluation; the ARD initial and final prior updates and final native evaluation are not included in the listed training/Jacobian intervals. Jacobian timing pertains to 100 validation examples and the particular architecture. Missing reference, anchor, transfer and evaluation components preclude interpreting these numbers as complete selector costs or combining them with historical CPU ID time.

Table A21. Logged duration components for new GPU pruning runs, seconds, mean $\pm$ sample SD over three seeds. Training loop includes periodic train/validation evaluations. ARM post-evaluation covers final native evaluations; ARD did not separately time them (NR). Anchor, transferred-candidate and other missing costs are not included here. These intervals are not complete end-to-end selector costs.

Dataset	Implementation	Training loop	Jacobian	Post-evaluation
MNIST	ARM	197.8 $\pm$ 2.2	—	0.043 $\pm$ 0.000
MNIST	ARD, zero	87.3 $\pm$ 0.3	0.011 $\pm$ 0.000	NR
MNIST	ARD, sample	87.1 $\pm$ 0.7	0.012 $\pm$ 0.002	NR
Fashion-MNIST	ARM	197.8 $\pm$ 1.5	—	0.043 $\pm$ 0.000
Fashion-MNIST	ARD, zero	88.2 $\pm$ 0.5	0.011 $\pm$ 0.000	NR
Fashion-MNIST	ARD, sample	88.0 $\pm$ 0.3	0.012 $\pm$ 0.002	NR
dSprites	ARM	664.8 $\pm$ 0.3	—	1.264 $\pm$ 0.001
dSprites	ARD, zero	299.2 $\pm$ 0.0	10.830 $\pm$ 0.169	NR
dSprites	ARD, sample	299.2 $\pm$ 0.2	11.047 $\pm$ 0.712	NR
CIFAR-10	ARM	328.4 $\pm$ 0.3	—	0.570 $\pm$ 0.001
CIFAR-10	ARD, zero	146.2 $\pm$ 0.1	6.968 $\pm$ 0.452	NR
CIFAR-10	ARD, sample	146.1 $\pm$ 0.1	7.505 $\pm$ 0.171	NR

The recorded `elbo` and `val_elbo` fields remain in the unmodified raw files for provenance but are not used as native ELBO measurements (Section 5.5). Additional pruning checkpoints were not saved, so native post-pruning ARD reconstruction or a corrected native likelihood cannot be recovered from these summary files alone. The analytic ARM-gradient check validates a component; it cannot replace those model-level evaluations.

Appendix G. Reproducibility Inventory and Remaining Records

The reaggregation entry point is `src/make51/rebuild_review.py`; it reads saved JSON and configuration files and writes numerical tables, six figures and the paired predictive analysis from the historical records. The new search analysis is `src/make51/rep`

lay_search.py; the bounded model validation is src/make51/validate_models.py. Its outputs are summarized by src/make51/summarize_validation.py. The additional-run reaggregation is src/make52/summarize_revision.py, and manuscript assembly is src/make52/build_manuscript.py. The generated main TeX embeds its table contents and references the figure PDFs. Compile from the project root with pdflatex repeatedly until references settle. The accompanying MAKE52_reproducibility.zip supplies the files, not only this inventory. Its README distinguishes inexpensive reaggregation from optional retraining, gives dependency versions and lists unavailable historical artifacts. MANIFEST.sha256 identifies the included payload. The archive contains a fixed source snapshot and is usable without access to a private repository.

Table A22. Source-to-output map for the present revision. Paths below are relative to the project root.

Evidence	File or directory
Configuration	config/experiment_config_MAKE49.json
Ordinary method records	results/make49/stage4_methods.json
Pruning records	results/make49/stage7_b7b8.json
Candidate metrics and latent variances	results/make49/stage4_cache/
Train/validation and readout trajectories	results/make49/stage3_e7a_e2.json
ID representation diagnostics	results/make49/stage5_e3_reference.json
Original AU prediction result	results/make49/stage5_e4_au_value.json
Synthetic runs	results/make49/stage6_e6_synthetic.json
All audited method rows	results/make51/method_results.csv
Paired AU analysis	results/make51/au_paired.json
Search replay and sensitivity	results/make51/search_replay.csv; results/make51/search_sensitivity.csv
Timing lineage	results/make51/cost_lineage.csv
Separate MNIST model records	results/make51/model_validation/
Additional start-width control	results/make52/start_width_ablation.json
Additional ARM/ARD records	results/make52/geco_arm_all.json; results/make52/ard_nactive_all.json
Additional-run aggregation	results/make52/manuscript_summary.json
Additional source identities	results/make52/manuscript_source_manifest.json
Snapshot identities	results/make51/source_manifest.json
Figures	results/figures/MAKE51/; results/figures/MAKE52/

Method-level row identifiers are (dataset, method, candidate seed, β or context). Candidate keys also contain width, requested epochs and stopping option. Every configuration/dataset–method–context cell in the main method archive has five seeds; this is checked during reaggregation. The ARD repetition does not create five distinct selection-weight conditions. Individual early off-grid cache entries are not all retained, although their final metrics and training traces remain in the method records. The original test predictions, historical full model checkpoints and independent ID/evaluation timings are not reconstructed from missing files.

The local change record documents the initial specification on 12 September; the switch to Bernoulli dSprites after collapse diagnostics; the move of ID estimation from an impossible 10,000-example validation sample to training data; trajectory runs without early stopping; and the 13 September reference-width, run-accounting and degenerate-output clarifications. The random-feature control was added after the primary predictive analysis. In this revision, anchor-cost completion, paired AU inference, quality-margin rereading and explicit implementation auditing are retrospective. None is relabelled as a publicly preregistered test. The 19 September local policies, threshold factorial, normalized ARD readout and new one-seed validation were specified after those earlier results. The supplement preserves their distinct specifications and output locations.

The additional start-width and pruning records are separate retrospective analyses. Reaggregation checks merged versus per-dataset files, three-seed coverage, 300-epoch curve

endpoints, raw count readouts, anchor targets and transferred cache metrics. This verifies consistency of the supplied records, not rerun equivalence of unavailable checkpoints.

References

- Kingma, D.P.; Welling, M. Auto-encoding variational Bayes. In *Proceedings of the 2nd International Conference on Learning Representations (ICLR)*, Banff, AB, Canada, 14–16 April 2014. arXiv:1312.6114. <https://doi.org/10.48550/arXiv.1312.6114>.
- Facco, E.; d’Errico, M.; Rodriguez, A.; Laio, A. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Sci. Rep.* **2017**, *7*, 12140. <https://doi.org/10.1038/s41598-017-11873-y>.
- Levina, E.; Bickel, P.J. Maximum likelihood estimation of intrinsic dimension. In *Advances in Neural Information Processing Systems*, **2004**, *17*. <https://proceedings.neurips.cc/paper/2004/hash/74934548253bcab8490ebd74afed7031-Abstract.html>.
- Bonheme, L.; Grzes, M. FONDUE: An algorithm to find the optimal dimensionality of the latent representations of variational autoencoders. arXiv **2022**, arXiv:2209.12806. <https://doi.org/10.48550/arXiv.2209.12806>.
- Burda, Y.; Grosse, R.; Salakhutdinov, R. Importance weighted autoencoders. In *Proceedings of the 4th International Conference on Learning Representations (ICLR)*, San Juan, Puerto Rico, 2–4 May 2016. arXiv:1509.00519. <https://doi.org/10.48550/arXiv.1509.00519>.
- Bonheme, L.; Grzes, M. Be more active! Understanding the differences between mean and sampled representations of variational autoencoders. *J. Mach. Learn. Res.* **2023**, *24*(324), 1–30. <https://jmlr.org/papers/v24/21-1145.html>.
- De Boom, C.; Wauthier, S.; Verbelen, T.; Dhoedt, B. Dynamic narrowing of VAE bottlenecks using GECO and L_0 regularization. arXiv **2020**, arXiv:2003.10901. <https://doi.org/10.48550/arXiv.2003.10901>.
- Saha, S.; Joshi, S.; Whitaker, R. ARD-VAE: A statistical formulation to find the relevant latent dimensions of variational autoencoders. arXiv **2025**, arXiv:2501.10901. <https://doi.org/10.48550/arXiv.2501.10901>.
- Tipping, M.E.; Bishop, C.M. Probabilistic principal component analysis. *J. R. Stat. Soc. Ser. B* **1999**, *61*, 611–622. <https://doi.org/10.1111/1467-9868.00196>.
- Tenenbaum, J.B.; de Silva, V.; Langford, J.C. A global geometric framework for nonlinear dimensionality reduction. *Science* **2000**, *290*, 2319–2323. <https://doi.org/10.1126/science.290.5500.2319>.
- Roweis, S.T.; Saul, L.K. Nonlinear dimensionality reduction by locally linear embedding. *Science* **2000**, *290*, 2323–2326. <https://doi.org/10.1126/science.290.5500.2323>.
- Ceruti, C.; Bassis, S.; Rozza, A.; Lombardi, G.; Casiraghi, E.; Campadelli, P. DANCo: An intrinsic dimensionality estimator exploiting angle and norm concentration. *Pattern Recognit.* **2014**, *47*, 2569–2581. <https://doi.org/10.1016/j.patcog.2014.02.013>.
- Johnsson, K.; Soneson, C.; Fontes, M. Low bias local intrinsic dimension estimation from expected simplex skewness. *IEEE Trans. Pattern Anal. Mach. Intell.* **2015**, *37*, 196–202. <https://doi.org/10.1109/TPAMI.2014.2343220>.
- Costa, J.A.; Hero, A.O. Geodesic entropic graphs for dimension and entropy estimation in manifold learning. *IEEE Trans. Signal Process.* **2004**, *52*, 2210–2221. <https://doi.org/10.1109/TSP.2004.831130>.
- Camastra, F.; Staiano, A. Intrinsic dimension estimation: Advances and open problems. *Inf. Sci.* **2016**, *328*, 26–41. <https://doi.org/10.1016/j.ins.2015.08.029>.
- Allegra, M.; Facco, E.; Denti, F.; Laio, A.; Mira, A. Data segmentation based on the local intrinsic dimension. *Sci. Rep.* **2020**, *10*, 16449. <https://doi.org/10.1038/s41598-020-72222-0>.
- Gałka, Ł.; Karczmarek, P.; Tokovarov, M. Isolation Forest Based on Minimal Spanning Tree. *IEEE Access* **2022**, *10*, 74175–74186. <https://doi.org/10.1109/ACCESS.2022.3190505>.
- Ansuini, A.; Laio, A.; Macke, J.H.; Zoccolan, D. Intrinsic dimension of data representations in deep neural networks. In *NeurIPS 2019*, Vol. 32, 2019. https://proceedings.neurips.cc/paper_files/paper/2019/file/cfcce0621b49c983991ead4c3d4d3b6b-Paper.pdf.

19. Pope, P.; Zhu, C.; Abdelkader, A.; Goldblum, M.; Goldstein, T. The intrinsic dimension of images and its impact on learning. In *ICLR 2021*, 2021. arXiv:2104.08894. <https://doi.org/10.48550/arXiv.2104.08894>.
20. Rezende, D.J.; Viola, F. Taming VAEs. arXiv **2018**, arXiv:1810.00597. <https://doi.org/10.48550/arXiv.1810.00597>.
21. Louizos, C.; Welling, M.; Kingma, D.P. Learning sparse neural networks through L_0 regularization. In *Proceedings of the 6th International Conference on Learning Representations (ICLR)*, Vancouver, BC, Canada, 30 April–3 May 2018. arXiv:1712.01312. <https://doi.org/10.48550/arXiv.1712.01312>.
22. Obata, C.; Jin'no, K. Analysis of the relationship between manifold structure and latent space capacity of input data using VAE. In *Proceedings of the 2025 International Symposium on Nonlinear Theory and Its Applications (NOLTA2025)*, Okinawa, Japan, 27–31 October 2025; pp. 32–35. Publication record: https://www.comm.tcu.ac.jp/nel/list_j_2025.html (accessed on 18 September 2026).
23. Whitney, H. The self-intersections of a smooth n -manifold in $2n$ -space. *Ann. Math.* **1944**, *45*, 220–246. <https://doi.org/10.2307/1969265>.
24. Lucas, J.; Tucker, G.; Grosse, R.B.; Norouzi, M. Don't blame the ELBO! A linear VAE perspective on posterior collapse. In *NeurIPS 2019*, Vol. 32, 2019. <https://proceedings.neurips.cc/paper/2019/file/7e3315fe390974fcf25e44a9445bd821-Paper.pdf>.
25. Rolinek, M.; Zietlow, D.; Martius, G. Variational autoencoders pursue PCA directions (by accident). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Long Beach, CA, USA, 15–20 June 2019; pp. 12406–12415. <https://doi.org/10.1109/CVPR.2019.01269>.
26. LeCun, Y.; Bottou, L.; Bengio, Y.; Haffner, P. Gradient-based learning applied to document recognition. *Proc. IEEE* **1998**, *86*, 2278–2324. <https://doi.org/10.1109/5.726791>.
27. Xiao, H.; Rasul, K.; Vollgraf, R. Fashion-MNIST: A Novel Image Dataset for Benchmarking Machine Learning Algorithms. *arXiv* **2017**, 1708.07747. <https://doi.org/10.48550/arXiv.1708.07747>.
28. Krizhevsky, A. Learning multiple layers of features from tiny images. *Technical Report*; University of Toronto: Toronto, ON, Canada, 2009.
29. Matthey, L.; Higgins, I.; Hassabis, D.; Lerchner, A. dSprites: Disentanglement testing sprites dataset. 2017. Available online: <https://github.com/google-deepmind/dsprites-dataset> (accessed on 12 September 2026).
30. Fu, H.; Li, C.; Liu, X.; Gao, J.; Celikyilmaz, A.; Carin, L. Cyclical annealing schedule: A simple approach to mitigating KL vanishing. In *Proceedings of NAACL-HLT 2019*, Minneapolis, MN, USA, 2019; pp. 240–250. <https://doi.org/10.18653/v1/N19-1021>.